

An advanced hyperheuristic approach for the home health care scheduling problem with time window in deterministic and uncertain environments

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ABSTRACT

In the last two decades, Home Health Care (HHC) services have garnered substantial attention from global researchers and practitioners. These services face significant challenges related to strategic, tactical, and operational decisions that must be solved for enhancing their efficiency and quality. Notably, researchers have focused significantly on the scheduling and routing aspects, recognizing their potential to enhance patient satisfaction, safety, and cost-effectiveness for both patients and HHC companies. Recent studies have highlighted the stochastic nature of these services, including random service and travel times, that may affect the scheduling and routing components. Nonetheless, a significant portion of recent research predominantly focuses on the deterministic version of the problem, disregarding its fundamental stochastic nature as emphasized by feedback from HHC companies. Within this context, we propose a two-stage mixed integer mathematical formulation to address the Home Health Care Scheduling and Routing Problem (HHCSR), encompassing stochastic demand, service, and travel times. Given the inherent Nondeterministic Polynomial (NP)-hardness of these scheduling and routing problems, we introduce a hyperheuristic approach based on a TS metaheuristic and Variable Neighborhood Search (VNS) heuristic. Extensive experiments validate the effectiveness of our proposed modeling and solving approaches, outperforming similar methods across various metrics. Notably, our stochastic model exhibits superior performance in terms of parameters like the number of patients served within time windows and computational time required to achieve optimal solutions. This study thus contributes a valuable decision support system for Home Health Care companies, empowering them to make informed decisions and construct robust schedules for their caregivers.

1. Introduction

In recent times, the home healthcare sector has witnessed rapid global growth across practical and research spheres. In practical terms, the escalation of home healthcare services worldwide is driven by the aging global population and the economic pressure exerted by these services on the profitability of home healthcare companies. As an illustration, the revenue of the home care services industry is projected at \$96.9B, with over 35,000 HHC companies serving 15 million patients and facilitating more than 600 million visits annually (Ankota, 2022). Simultaneously, the research landscape has seen the emergence of numerous studies published in reputable journals, presenting solutions to the Home Health Care Scheduling and Routing Problem (HHCSR). For

instance, Di Mascolo et al. (2021) conducted a comprehensive literature survey on routing and scheduling in home health care, reviewing over 50 publications from the recent years, indicating a robust interest in this domain.

However, a notable portion of the literature concentrates on the deterministic version of the issue, accommodating solely deterministic scenarios and disregarding the problem's inherent stochastic nature. This trend is evident in works like those of Chaieb, Jemai et al. (2022) and Dessevre et al. (2022). However, as underlined by various home healthcare companies, it is essential for models to consider stochastic decisions while devising specific schedules for customer visits, as cited by Nikzad et al. (2021).

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In the realm of healthcare services, numerous scenarios arise as staff teams provide care to patients. Uncertainties in travel time, stemming from factors like weather conditions, peak hours, accidents, vehicle availability, and road conditions, contribute to this complexity (Chaieb, Sassi et al., 2022; Stratman et al., 2023). Additionally, service times for each patient can vary due to factors such as the complexity of the case and the duration of diagnosis.

Moreover, the demand for healthcare services is not constant, thanks to variables like visit cancellations, appointment changes, real-time urgent requests, and the need for service synchronization. This variability introduces a stochastic dimension to the Home Health Care Scheduling and Routing Problem (HHCSRP), making it crucial to consider these uncertainties. Neglecting these variables can lead to suboptimal or even infeasible solutions, complicating the problem further.

In real-world scenarios, certain parameters, including decision variables, exhibit stochastic behavior due to uncertainties in customer service times, service processes, and more. This stochasticity is pervasive in various problem domains, including home healthcare services, where tardiness at patients' homes incurs penalties in the overall cost. For example, caregivers' arrival times at patients' homes may deviate from the scheduled appointment times due to stochastic travel times. Furthermore, service times often exhibit variability due to differences in caregiver and patient conditions. For instance, if caregivers arrive late, patients may experience delays, leading to additional costs for the home healthcare company, such as managing patient inconvenience and overtime expenses resulting from violating nurses' standard working hours. When these uncertainties coincide, the scheduling and routing process becomes even more intricate, as decisions made at one point in time ripple through subsequent periods. Robust decision-making becomes essential to anticipate disruptions.

This complexity necessitates the use of stochastic modeling techniques, which leverage various methods to assess the probability of different outcomes using random variables under varying conditions. Such modeling techniques are widely adopted across industries worldwide. Companies employ stochastic modeling to efficiently manage their assets, optimize business practices, and enhance overall profitability.

This study strives to introduce a streamlined solution for the Home Health Care Scheduling and Routing Problem (HHCSRP) that accounts for real-life home health care (HHC) scenarios involving uncertainty. To this end, we establish an integrated two-level stochastic model to address the problem comprehensively. The first stage entails scheduling patient visits, modeled as a nurse rostering problem recognized as NP-hard in prior work (Burke et al., 2004). In the second stage, we map out staff routes to reach patients' residences, treating this element as a Vehicle Routing Problem with Time Windows (VRPTW), also acknowledged as NP-hard (Bräysy & Gendreau, 2005). Consequently, the composite problem is inherently NP-hard, combining two NP-hard constituents.

The model uniquely considers three stochastic variables: (i) demand service, (ii) travel time, and (iii) service time. Significantly, no existing research has examined the HHCSRP with stochastic factors in all three factors, presenting a formidable challenge in achieving optimal or near-optimal solutions necessitating expected value assessment. Conventional solving techniques, such as commercial solvers, exhibit limitations in addressing large-scale instances of NP-hard problems with stochastic components, as noted in Tahami and Fakhraei (2022).

Due to the inherent NP-hardness of the problem, we propose employing a Tabu Search metaheuristic, augmented with a Variable Neighborhood Search (VNS) heuristic, to tackle the challenge. The synergy of these methods aims to enhance algorithmic performance. Through systematic experimental studies and comparative analysis, we validate the efficiency and effectiveness of the proposed modeling and solving approach.

In conclusion, the contributions of this paper are three-fold:

- We introduce a two-stage mixed integer mathematical formulation to effectively model the HHCSRP.
- Utilizing uncertainty theory, we address uncertain demand, service, and travel times. Notably, this study marks the first instance of a collective examination of these three variables of uncertainty.
- We present an efficient and robust metaheuristic algorithm based on Tabu Search (TS), embedded by Variable Neighborhood Search (VNS), adeptly managing practical and sizable problem instances. A comparative study to assess our solution compared to similar approaches.

Additionally, a comprehensive comparative study is undertaken to gauge the performance of our solution compared to analogous methodologies.

The remainder of the work will be organized as follows. In Section 2, we investigate the literature related to the HHCSRP and the stochastic theory. In Section 3, we model the HHCSRP as a two-stage mixed integer mathematical model. To solve the problem efficiently, we illustrate the TS-based metaheuristic algorithm integrated with the VNS heuristic in Section 4. The computational experiments are conducted and the analysis study is presented in Section 5. Section 6 summarizes this study with some conclusions and future research directions.

2. Literature review

Research on the Home Health Care Scheduling and Routing Problem (HHCSRP) with stochastic demand, service, and travel times is limited. While specific investigations have examined stochastic aspects like service time or travel time individually, a comprehensive exploration of the full stochastic HHCSRP in a general context is lacking. Although existing studies tackle broader questions related to home healthcare services, they do not specifically address the complete stochastic version of HHCSRP. This section is devoted to discuss the stochastic modeling, exploring its role in problem-solving. The exploration begins by defining stochastic modeling and introducing key terminology. Recent studies focused on the Home Health Care Scheduling Problem with Time Window in uncertain environments will be reviewed. This knowledge will be synthesized to tackle the challenging HHCSRP, combining scheduling and routing aspects in a complex setting.

The following topics will be covered in this section:

- Terminology
- Related Works
- Literature Findings

2.1. Terminology

In referencing the terminology utilized within the literature to convey a description to the HHCSRP, it is important to clarify that these terminologies are not employed as methodologies in those works. Instead, they are cited to illustrate a synonymous meaning for enhanced contextual understanding of the Home Health Care Scheduling and Routing Problem. In this context, different terminologies were used to name this kind of HHCSRP which we cite in the following:

- **Stochastic:** refers to a process or system that involves randomness or probability and the outcomes are not determined with certainty but are instead subject to varying probabilities. The term was used in the works of Carello and Lanzarone (2021), Chen et al. (2021), Du et al. (2019), Goodarzi et al. (2021), Nikzad et al. (2021) and Zheng et al. (2021).
- **Uncertainty:** refers to a lack of sureness or predictability about the outcomes of a situation or event. It indicates that the future state is not precisely known due to incomplete information or varying factors. The term was used in the works of Bazirha et al. (2021), Carello and Lanzarone (2021), Chen et al. (2021), Du et al. (2019), Gilmour et al. (2017), Nikzad et al. (2021) and Zheng et al. (2021).

- **Real-time:** pertains to actions that process and respond to data with minimal delay, enabling rapid reactions to changing conditions. The term was used in the works of Du et al. (2019), Goodarzian et al. (2021), Nikzad et al. (2021), Sandhu and Sood (2017) and Yuan et al. (2018).
- **Dynamic:** in which the elements or parameters evolve over time, often influenced by external factors, interactions, or conditions. The term was cited in the works of Carello and Lanzarone (2021), Du et al. (2019), Goodarzian et al. (2021) and Sandhu and Sood (2017).
- **Non-deterministic:** refers to situations where the outcome is not determined solely by the initial conditions or inputs. Randomness or uncertainty can lead to different results. The term was cited in the works of Liu et al. (2021), Pillac et al. (2013), Shahnejat et al. (2021) and Shi et al. (2017, 2019).
- **Probabilistic:** in which the events or results are described in terms of the likelihood of their occurrence, rather than being fixed or certain. The term was cited in the works of Bazirha et al. (2021), Campbell and Thomas (2008), Fang (2023) and Zhan and Wan (2018).

2.2. Related works

The rapid growth of HHC services motivates both researchers and practitioners to study the issues related to both scheduling and routing of home health caregivers. As mentioned above, the set of works studying the stochastic version of the HHCSR is reduced compared to those studying the deterministic variant of the problem. In this section, we survey the works concerning the HHCSR with stochastic demand, i.e. uncertain travel and services times.

The rapid growth of HHC services serves as a catalyst for both researchers and practitioners to study and handle the scheduling and routing for home health caregivers. As highlighted earlier, the pool of studies examining the stochastic version of the Home Health Care Scheduling and Routing Problem (HHCSR) is notably smaller in comparison to those addressing the deterministic counterpart. In this section, we conduct an overview of research centered around the HHCSR involving stochastic demand, encompassing uncertainties in travel and service times.

Zhan et al. (2021) addressed the integration of routing and appointment scheduling problems with stochastic service times. They proposed a mixed-integer program solved using an L-shaped method. For large problems, a heuristic was developed for efficient problem solving. Additionally, two heuristics were introduced: one for the Traveling Salesman Problem without appointment scheduling, and an OVP heuristic that generates scheduling sequences while omitting routing decisions. Experimental validation employed randomly generated data instances, demonstrating efficient performance compared to the CPLEX solver across most cases. However, a notable limitation lies in the exclusion of random traveling distances, which could potentially impact the overall validity of the study.

Shi et al. (2019) tackled the stochastic HHCSR considering uncertain travel and service times. They introduced a Robust Optimization (RO) model utilizing budget uncertainty theory and a complex recursive function for nurses' arrival times. Two solving approaches were suggested: an exact method (Gurobi solver) and three metaheuristics (Tabu Search, Variable Neighborhood Search, Simulated Annealing (SA)). Experimental comparison highlighted the robustness of their RO model, even when accounting for uncertainties.

Shi et al. (2017) addressed HHCSR with time windows and fuzzy demand using a fuzzy chance constraint-programming model. Hybrid Genetic Algorithm (GA) combined with stochastic simulation was employed. Experiments on benchmark instances demonstrated the effectiveness of their approach, despite limitations in addressing only fuzzy demand.

Zheng et al. (2021) discussed service authorization and capacity planning in HHCSR with stochastic demand. A nested decomposition approach led to a two-level stochastic programming model, focusing on capacity planning, service authorization, and service hour assignment. A supergradient based algorithm showcased effective performance, albeit with a focus on fuzzy demand.

Yang et al. (2021) treated HHCSR as a multi-objective problem with uncertain travel and service times. They introduced an uncertain programming model and proposed a Multi-Objective Artificial Bee Colony metaheuristic. Experiments demonstrated efficiency in balancing workload and costs.

Zhan and Wan (2018) investigated scheduling tasks in-home services, employing an appointment and routing scheduling with team assignment model. A Tabu Search-based heuristic algorithm was devised. This comprehensive approach considered team assignment, routing, and appointment scheduling simultaneously, with random service times.

Restrepo et al. (2020) studied integrated staffing and scheduling in-home healthcare. Their two-stage stochastic programming model optimized staff allocation, considering over/under-covering costs. Experimental results showcased effective utilization, reduced costs, and enhanced service levels.

Shahnejat et al. (2021) addressed HHCSR with temporal dependencies under uncertainty, offering a model considering various constraints. Metaheuristic algorithms (GA, memetic algorithm, SA) were proposed, with the memetic algorithm outperforming. Dynamic model and dispatching policies were also explored.

Nikzad et al. (2021) focused on home healthcare human resource planning, utilizing a two-stage stochastic mixed integer model. A four-phase algorithm and metaheuristic-based methods were proposed for solving. The model excelled in solving large instances, albeit with certain deterministic assumptions.

Bazirha et al. (2021) considered uncertainty in both service and travel times, with synchronization of services. A two-level stochastic programming model addressed transportation costs and expected value of recourse. The model was solved using metaheuristic algorithms and dynamic dispatching policies.

Liu et al. (2019) examined home-caregiver scheduling and routing with stochastic travel and service times. A route-based mathematical model was proposed, solved using a combined approach. The approach showed promise, but further validation through various solving techniques was suggested.

Du et al. (2019) investigated real-time scheduling optimization in-home healthcare services. An effective real-time scheduling model and an improved memetic algorithm were suggested. Experiments on real-life data instances showcased the model's efficiency in real-time scenarios.

Tsang and Shehadeh (2023) discussed the routing and appointment scheduling problem with uncertain travel and service times arising from home service practice. To solve the problem, the authors suggested a stochastic programming model and two distributionally RO models. In all models, authors assumed known and unknown travel and service time distributions with two types of ambiguity sets for the distributionally Robust Models which are the 1-Wasserstein ambiguity and the mean-support sets.

Bazirha et al. (2023) proposed a Stochastic Programming model with Recourse (SPR) to handle the HHCSR considering the uncertainties of traveling and caring times. They aimed to minimize the expected value of recourse and the transportation cost. The value of recourse was defined as a remuneration for nurses' extra working time and a penalty cost for patients' delayed services. They solved the deterministic model by CPLEX, GA and VNS while they used a Monte Carlo simulation embedded into the GA to solve the SPR model. The experimental study proves the efficiency of the VNS and the GA and the complexity of the SPR model in terms of Central Processing Unit (CPU) running times.

Table 1
Stochastic variables considered in the literature.

Study	Stochastic travel time	Stochastic service time	Stochastic demand
Zhan et al. (2021)		✓	
Shi et al. (2019)	✓	✓	
Shi et al. (2017)			✓
Zheng et al. (2021)			✓
Yang et al. (2021)			✓
Zhan and Wan (2018)		✓	
Restrepo et al. (2020)			✓
Shahnejat et al. (2021)	✓	✓	
Nikzad et al. (2021)	✓	✓	
Bazirha et al. (2021)	✓	✓	
Liu et al. (2019)	✓	✓	
Du et al. (2019)			✓
Bazirha et al. (2023)	✓	✓	
Tsang and Shehadeh (2023)	✓	✓	
Naderi et al. (2023)	✓	✓	
Our study	✓	✓	✓

Naderi et al. (2023) studied the multi-period planning and scheduling of home-care services considering the uncertainty in service and travel times. To model the problem, the authors suggested a mixed-integer deterministic and robust optimization models. To efficiently solve both models, the authors used the same novel exact technique that demonstrate that the fully robust solution increases the mean cost of nominal solutions by 18%. Moreover, the authors confirmed that cost-savings of caregiver flexibility was about 19%.

These studies contribute to understanding different facets of the HHCSR, including handling uncertainty, dynamic aspects, multiple objectives, and varying constraints. While each study has unique strengths, certain limitations, such as focus on specific uncertain variables or deterministic assumptions, are shared across multiple works.

2.3. Literature review findings and the study contribution

Numerous scholarly articles within the literature study the Home Health Care Scheduling Problem with Time Window under conditions of uncertainty. The extensive attention given to this subject underscores its significance to both Home Health Care (HHC) companies and patients, as it facilitates cost-efficiency and attaining health-related goals.

In this context, each examined study presents empirical investigations showcasing advancements in both the scheduling and routing of HHC services. Several decision support tools have been proposed, aimed at aiding decision-makers in formulating superior and optimal scheduling and routing strategies. Additionally, multiple studies underscore the foundational role of uncertainty when addressing real-world HHCSRPs. However, it is noteworthy that each studied paper focuses on distinct stochastic variables. To provide a succinct overview, we present in the following table the specific stochastic variables considered within each of the reviewed works (see Table 1).

In our investigation, we address the HHCSR incorporating a Time Window under an environment of uncertainty. This approach encompasses travel, distance, and stochastic demand considerations, which, as far as our knowledge extends, have not been explored in existing literature. Remarkably, our study pioneers an attempt to tackle this uncharted terrain.

Furthermore, our approach introduces a Tabu Search metaheuristic that embeds a VNS heuristic to effectively manage this multifaceted problem.

3. Problem description and mathematical formulation

The HHCSR is a difficult process due to the high number of constraints to be satisfied which may be sometimes conflicting. In

addition, varying practical conditions may increase the complexity of the problem. In this part, we go over detailing the main objectives to realize when solving the HHCSR by providing a detailed description. Subsequently, we proceed to outline a problem formulation by constructing a mathematical model.

3.1. Problem description

The Home Health Care Scheduling and Routing Problem represents a challenging combinatorial optimization task that revolves around delivering intensive care to patients within their homes. In essence, the problem encapsulates the following scenario: a designated group of p patients requires medical attention in their homes, attended to by n medical personnel. These caregivers initiate their routes from a central depot (the service center), visiting geographically dispersed clients, and ultimately returning to the central depot upon completing their rounds. On the other hand, certain patients necessitate laboratory services, prompting caregivers to visit the laboratory node for either delivery or pickup services.

The process has the following basic assumptions which we adopted from Shahnejat et al. (2021), the work we will be comparing our results to, ensuring a meaningful and relevant comparison:

- Each patient can request care within a known time window.
- Nurses have a maximum working time.
- Visits can be performed by only qualified nurses (with the required skills).
- Service time required to provide care to a patient is deterministic or probabilistic.
- Travel time taken to travel between different patients' locations is deterministic or probabilistic.
- Geographical locations of patients and caregivers are known, and the distances or travel times between these locations can be calculated.

The HHCSR aims mainly to optimize two tasks which are (i) the scheduling of nurses to patients and (ii) the routing of caregivers to customers' homes. Whether in the first or the second goal, changes may occur in real-time processing that may affect the general alignment. These changes may occur due to many unexpected events disarranging the established scheduled plan which requires readjusting of the schedule and the routes. In Fig. 1, we present the different situations that may occur and results in a stochastic version of the problem.

3.2. Problem formulation

In this section, we built the mathematical model for the HHCSR with a Time window under uncertainty by employing fuzzy chance-constrained programming. So, we will first present the mathematical notations of the model to clearly describe the model. To do so, we adopted the mathematical model presented in Shahnejat et al. (2021) which considers the stochastic service and travel time. Besides, we considered the stochastic demand in the model since the service demand is generally imprecise in real-world decision-making problems. In the following, we will present the set of notations, then we present the mathematical model composed of the objective functions to realize, and the set of constraints to satisfy.

3.2.1. Notations

In Table 2, we highlight the required notations for building the mathematical model by presenting the indices, the sets, the parameters, and the decision variables.

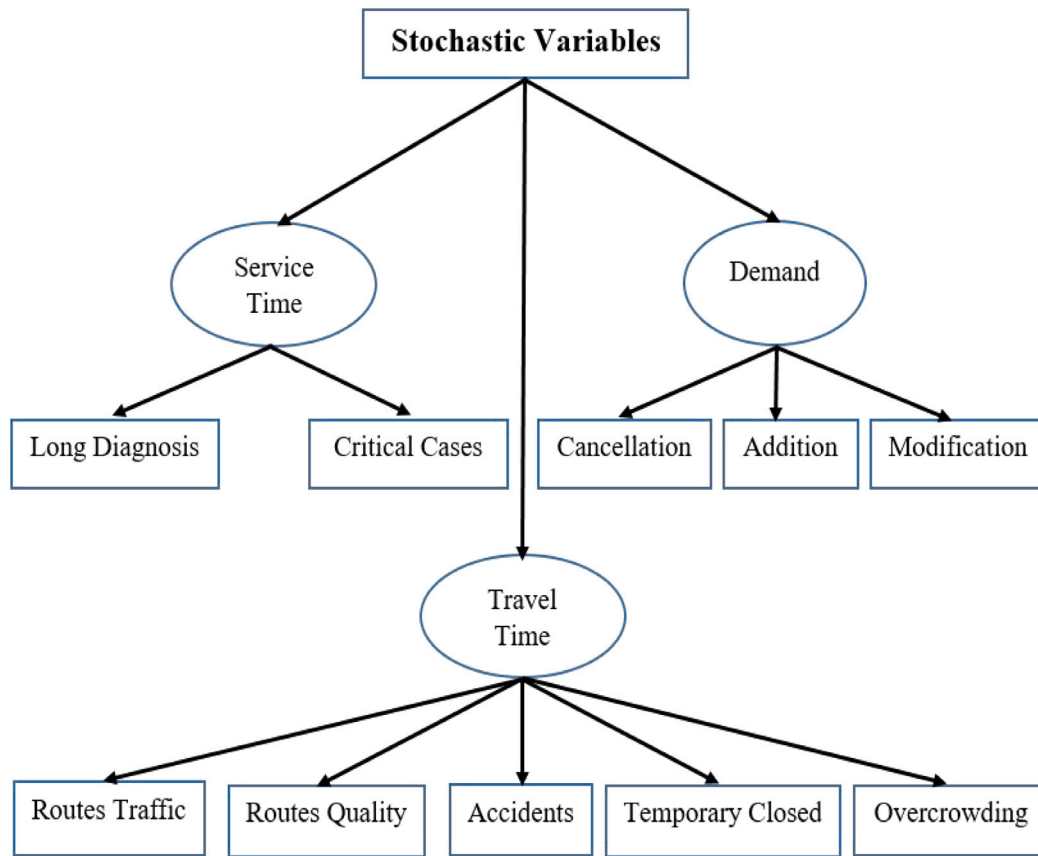


Fig. 1. Stochastic variables influencing the home health care scheduling and routing problem.

Table 2

Problem notations.

Indices	Description
k	Index of the set of nurses
s	Index of the set of services
t, r	Index of periods of time during a working day
i, j	Index of all nodes including the starting and ending nodes, the index of the central depot, and the index of all nodes
d	Index of the starting node
f	Index of the ending node
l	Index of the laboratory.
Sets	Description
P	Set of all nodes along with the patients the laboratory, and the central depot
N	Set of nurses
S	Set of services
S_{lab}	Set of laboratory services
T	Set of time periods during a working day
Parameters	Description
c_{ij}	Transportation cost to move from node i to node j .
t_{ijt}	Travel time needed to move from node i to node j at time period t
$[e_{it}, l_{it}]$	Service time window for the patient i .
st_{it}^k	Service time for serving the patient i at the time period t .
cs_{pk}	Cost of performing the service p by the nurse k .
s_{ikt}	Starting time of the nurse k to serve patient i at period t .
ft_{kt}	Maximum time allowed for returning nurse k at period t to be ready for the future time period
M	A very large positive number
Decision variables	Description
x_{ijkt}	1, if the nurse k travels from patient i (node) to the patient (node) j at period t , 0 otherwise.
d_{ipt}	1, if the patient i needs a service p at the time period t 0, otherwise
q_{pk}	1, if the nurse k has the required skills to perform the service p 0, otherwise

3.2.2. Mathematical model

In the following subsections, we present a mathematical formulation of the objective function and the set constraints to be respected when solving the model.

• Objective Function

We stress once again here that the main aim of solving the HHCRSP with a Time window is to minimize the operating costs. So, to achieve this aim, the model should realize the following objective function:

$$\text{Min} \sum_{i \in P} \sum_{j \in P} \sum_{k \in N} \sum_{t \in T} c_{ij} x_{ijkt} + \sum_{i \in P} \sum_{j \in P} \sum_{k \in N} \sum_{p \in S} \sum_{t \in T} c_{spk} q_{pk} x_{ijkt} \quad (1)$$

The first part of the objective function (1) aims to minimize the total traveling cost. While the second part minimizes the cost of ensuring the services.

• Constraints

To ensure that each patient is either served by a nurse or not handled at all, at each time period, the model must respect the following constraint:

$$\sum_{i \in P} \sum_{k \in N} x_{ijkt} \leq 1 \quad \forall j \in P, \forall t \in T \quad (2)$$

To ensure that each nurse start and finish their duty tour at the determined places, the model must respect the following constraint:

$$\sum_{j \in P} x_{djkt} \leq 1 \quad \forall k \in N, \forall t \in T \quad (3)$$

To ensure the flow conservation related to the laboratory node, the model must respect the two following constraints:

$$\sum_{i \in P} x_{ilkt} \leq 1 \quad \forall k \in N, \forall t \in T \quad (4)$$

$$x_{lftk} \leq 1 \quad \forall k \in N, \forall t \in T \quad (5)$$

To ensure the flow conservation related to the routes and their continuity, the model must respect the following two constraints:

$$\sum_{i \in P} x_{ihkt} = \sum_{j \in P} x_{hjkt} \quad \forall k \in N, \forall h \in T, \forall t \in T \quad (6)$$

$$\sum_{i \in P} x_{ilkt} = \sum_{j \in P} x_{ljkt} \quad \forall k \in N, \forall t \in T \quad (7)$$

To ensure the planning of the patients with demands and excludes patients with no need to care, the model must contain the following two constraints:

$$\sum_{i \in P} \sum_{k \in N} x_{ijkt} = \sum_{s \in S} d_{jpt} \quad \forall j \in P, \forall t \in T \quad (8)$$

$$\sum_{s \in S} d_{jpt} \leq M \sum_{i \in P} \sum_{k \in N} x_{ijkt} \quad \forall j \in P, \forall t \in T \quad (9)$$

To ensure that each patient is served by only qualified nurses based on his needs at each time period, the model must respect the following constraint:

$$x_{ijkt} - M(1 - d_{jpt}) \leq q_{pk} \quad \forall i, j \in P, \forall k \in N, \forall s \in S, \forall t \in T \quad (10)$$

To ensure that all patients must be visited within their time window, the model must respect the following constraint:

$$e_{jt} \sum_{i \in P} x_{ijkt} \leq s_{jkt} \leq l_{jt} \sum_{i \in P} x_{ijkt} \quad \forall j \in P, \forall k \in N, \forall t \in T \quad (11)$$

To ensure that the arrival and starting service time are respected for each patient at each time period and also the service consistency, the model must respect the following constraint:

$$s_{ikt} + d_{it}^k + t_{ijt} + s_{it}^k - M(1 - x_{ijkt}) \leq s_{jkt} \quad \forall i, j \in P, \forall k \in N, \forall s \in S, \forall t \in T \quad (12)$$

To ensure that the working hours of each nurse is respected to have enough time to provide efficient service for the next time period if the nurse has been assigned to serve patients at two consecutive time periods.

$$s_{lkt} - f_{tkt} - M(2 - av_{kt} + \sum_{i \in P} x_{ilkt}) \leq M(1 - x_{djkt+1}) \quad \forall j \in P, \forall k \in N, \forall t \in T \quad (13)$$

To ensure the continuity of care for the patients at different time periods in relation to the assignment procedure, the model must respect the following constraint:

$$1 - M(2 - (cc_{jtr} + \sum_{i \in P} x_{ijkt})) \leq \sum_{i \in P} x_{ijktr} \quad \forall j \in P, \forall t, r \in T, t < r, \forall k \in N \quad (14)$$

To ensure the selection of the nurses for ensuring the care in each period, the model must respect the following constraint:

$$\sum_{i \in P} \sum_{j \in P} x_{ijkt} \leq M q_{pk} \quad \forall k \in N, \forall t \in T, \quad (15)$$

To define the nature of the variables and their domain, the model must respect the following two constraints:

$$x_{ijkt}, q_{kt}, d_{ipt}, cc_{itr} \in \{0, 1\} \quad \forall i, j \in P, \forall k \in N, \forall s \in S, \forall t \in T \quad (16)$$

$$s_{ikt} \geq 0 \quad \forall i \in P, \forall k \in N, \forall t \in T \quad (17)$$

• Travel Time, Service Time and Demand Uncertainty

As mentioned above, real-world situations are often, contain some uncertain parameters such as the demand parameter, the travel and service times, etc. which may lead to a high degree of uncertainty. Therefore, considering the previous deterministic model of the HHCRSP with a Time window, which ignores the uncertainty features in parameters, may not be a suitable and reasonable choice. Moreover, unfeasible solution space or even non-optimal solutions may occur when ignoring uncertainty in the proposed mathematical model. To deal with such uncertainty, several techniques have been proposed in the literature.

In our context, we choose to develop a stochastic optimization model that was initially suggested by Ben-Tal and Nemirovski (1999) which can integrate decision-making risk attitudes when dealing with fuzzy scenarios. We used this technique as an appropriate choice for decision-making since it considers uncertain parameters. Thus, we developed an HHCRSP stochastic version with travel, service times, and demand uncertainty. To do so, stochastic optimization involves a clear and practical definition of the uncertainty sets to efficiently represent the uncertain parameters. Based on the above discussions and observations, we define three types of uncertainty sets for each caregiver used k :

- The customer demand uncertainty set U_d^k
- The travel time uncertainty set U_{it}^k
- The service time uncertainty set U_{st}^k

Based on the work of Hu et al. (2018), the demand uncertainty sets can be defined as follows:

$$\vartheta_d = X_{k \in K} U_d^k \quad (18)$$

Eq. (18) describes the overall demand uncertainty as the Cartesian product of uncertain demand service for the i th patient in route

k.

$$U_d^k = \{\tilde{d}_i \in R^{[A^{k,t}]} | \widehat{d_{ijt}} = \overline{d_{ijt}} + \gamma_{ijt} \widehat{d_{ijt}}, \sum_{i \in N^{k,t}} |\gamma_{ijt}| \leq r_d^{k,t}, |\gamma_{it}| \leq 1, r_d^{k,t} = [\theta_d | N^{k,t}|], \forall i \in N^{k,t}\} \quad (19)$$

Eq. (19) describes the customer demand uncertainty set U_d^k . $N^{k,t}$ presents the set of customers in a route to be satisfied by the nurse k. $\tilde{d}_i$ presents the uncertain demand for patient i, $\widehat{d_{ijt}}$ presents the maximum deviation of the demand uncertainty, and $\overline{d_{ijt}}$ presents the nominal demand for the patient i. $\tilde{d}_i$ has random value that exists in the interval $\overline{d_{ijt}} - \widehat{d_{ijt}}, \overline{d_{ijt}} + \widehat{d_{ijt}}$. The parameter γ_i presents an auxiliary variable. $r_d^{k,t}$ presents a variable from the budget uncertainty that monitors the uncertainty of the demand level. $\theta_d \in [0,1]$ presents a coefficient of the demand uncertainty. $[\theta_d | N^{k,t}|]$ presents the smallest integer value greater than the real value $\theta_d | N^{k,t}|$.

Based on the work of Hu et al. (2018), we define the overall travel time uncertainty sets as follows:

$$\theta_{it} = X_{k \in K} U_{it}^k \quad (20)$$

Eq. (20) describes the overall travel time uncertainty as the Cartesian product of uncertain service time for the i th caregiver in route k.

$$U_{it}^k = \{\tilde{t}_{it} \in R^{[N^{k,t}]} | \widehat{t_{ijt}} = \overline{t_{ijt}} + \alpha_{ijt} \widehat{t_{ijt}}, \sum_{(i,j) \in A^{k,t}} |\alpha_{ijt}| \leq r_t^{k,t}, |\gamma_{it}| \leq 1, r_t^{k,t} = [\theta_t | A^{k,t}|], \forall (i,j) \in A^{k,t}\} \quad (21)$$

Eq. (21) describes the service time uncertainty set U_{it}^k . $A^{k,t}$ presents the set of arcs in a route k. $\tilde{t}_{it}$ presents the uncertain travel time for patient i, $\widehat{t_{ijt}}$ presents the maximum deviation of the travel time uncertainty, and $\overline{t_{ijt}}$ presents the nominal travel time for the patient i. $\tilde{t}_{it}$ has random value that exists in the interval $\overline{t_{ijt}} - \widehat{t_{ijt}}, \overline{t_{ijt}} + \widehat{t_{ijt}}$. The parameter α_i presents an auxiliary variable. $r_t^{k,t}$ presents a variable from the budget uncertainty that monitors the uncertainty of the travel level. $\theta_t \in [0,1]$ presents a coefficient of the travel service uncertainty. $[\theta_t | N^{k,t}|]$ presents the smallest integer value greater than the real value $\theta_t | N^{k,t}|$. Similarly to the overall demand and travel time uncertainty sets, the service time can be defined as follows:

$$\theta_{st} = X_{k \in K} U_{st}^k \quad (22)$$

Eq. (22) describes the overall service time uncertainty as the Cartesian product of uncertain service time for the i th patient in route k.

$$U_{st}^k = \{\tilde{s}_{it} \in R^{[A^{k,t}]} | \widehat{s_{ijt}} = \overline{s_{ijt}} + \beta_{ijt} \widehat{s_{ijt}}, \sum_{(i,j) \in A^{k,t}} |\beta_{ijt}| \leq r_{st}^{k,t}, |\beta_{it}| \leq 1, r_{st}^{k,t} = [\theta_{st} | A^{k,t}|], \forall (i,j) \in N^{k,t}\} \quad (23)$$

Eq. (23) describes the service time uncertainty set U_{st}^k . $N^{k,t}$ presents the set of customers in a route to be satisfied by the nurse k. $\tilde{s}_{it}$ presents the uncertain service time for patient i, $\widehat{s_{ijt}}$ presents the maximum deviation of the service time uncertainty, and $\overline{s_{ijt}}$ presents the nominal service time for the patient i. $\tilde{s}_{it}$ has a random value that that exists in the interval $\overline{s_{ijt}} - \widehat{s_{ijt}}, \overline{s_{ijt}} + \widehat{s_{ijt}}$. The parameter β_i presents an auxiliary variable. $r_{st}^{k,t}$ presents a variable from the budget uncertainty that monitors the uncertainty of the service level. $\theta_{st} \in [0,1]$ presents a coefficient of the time service uncertainty. $[\theta_{st} | N^{k,t}|]$ presents the smallest integer value greater than the real value $\theta_{st} | N^{k,t}|$.

• Stochastic Optimization Model

To consider the above-defined stochastic sets, the deterministic model previously built will be extended to form a stochastic model by integrating constraints (24), (25), and (26) instead

of constraints (11), (12), and (13). To do so, we will adjust our optimization model by adapting some decision variables in the deterministic model. In our context, we will only adjust the demand vector $d \in ext(\theta_d)$, the travel time vector $tt \in ext(\theta_{tt})$, and the service time vector $st \in ext(\theta_{st})$. It is important to mention here that $ext(\theta_d)$, $ext(\theta_{tt})$, and $ext(\theta_{st})$ are sets that contain all the extreme values of sets θ_d , θ_{tt} , and θ_{st} respectively. Accordingly, each variable in the mathematical model is changed to S_{ikt} , where $S_{ikt}(d, tt, st)$ is a function of d, tt , and st according to Agra et al. (2013). Thus, the set of constraints to be adapted and injected into the new stochastic model can be formulated as follows:

$$e_{jt} \sum_{i \in P} x_{ijkt} \leq S_{ikt}(d, tt, st) \leq l_{jt} x_{ijkt} \quad \forall j \in P, \forall k \in N, \forall t \in T, \quad \forall d \in ext(\theta_d), \forall tt \in ext(\theta_{tt}), \forall st \in ext(\theta_{st}) \quad (24)$$

$$S_{ikt}(d, tt, st) + d_{it}^k + tt_{ijt} + st_{it}^k - M(1 - x_{ijkt}) \leq S_{jkt}(d, tt, st) \quad \forall i, j \in P, \forall k \in N, \forall t \in T, \forall d \in ext(\theta_d), \forall tt \in ext(\theta_{tt}), \forall st \in ext(\theta_{st}) \quad (25)$$

$$S_{ikt}(d, tt, st) - f_{tkt} - M(2 - av_{kt} + \sum_{i \in P} x_{ilkt}) \leq M(1 - x_{djk(t+1)}) \quad \forall j \in P, \forall k \in N, \forall t \in T, \forall d \in ext(\theta_d), \forall tt \in ext(\theta_{tt}), \forall st \in ext(\theta_{st}) \quad (26)$$

So, we can model the stochastic version of the problem, that encompasses all the referenced formulas from the preceding sections, as follows:

$$\begin{aligned} & Min \sum_{i \in P} \sum_{j \in P} \sum_{k \in N} \sum_{t \in T} c_{ij} x_{ijkt} + \sum_{i \in P} \sum_{j \in P} \sum_{k \in N} \sum_{p \in S} \sum_{t \in T} c_{spk} q_{pk} x_{ijkt} \\ & \sum_{i \in P} \sum_{k \in N} x_{ijkt} \leq 1 \quad \forall j \in P, \forall t \in T \\ & \sum_{j \in P} x_{djk(t)} \leq 1 \quad \forall k \in N, \forall t \in T \\ & \sum_{i \in P} x_{ilkt} \leq 1 \quad \forall k \in N, \forall t \in T \\ & x_{ifkt} \leq 1 \quad \forall k \in N, \forall t \in T \\ & \sum_{i \in P} x_{ihkt} = \sum_{j \in P} x_{hjkt} \quad \forall k \in N, \forall h \in T, \forall t \in T \\ & \sum_{i \in P} x_{ilkt} = \sum_{j \in P} x_{ljk(t)} \quad \forall k \in N, \forall t \in T \\ & \sum_{i \in P} \sum_{k \in N} x_{ijkt} = \sum_{s \in S} d_{jpt} \quad \forall j \in P, \forall t \in T \\ & \sum_{s \in S} d_{jpt} \leq M \sum_{i \in P} \sum_{k \in N} x_{ijkt} \quad \forall j \in P, \forall t \in T \\ & x_{ijkt} - M(1 - d_{jpt}) \leq q_{pk} \quad \forall i, j \in P, \forall k \in N, \forall s \in S, \forall t \in T \\ & 1 - M(2 - (cc_{jtr} + \sum_{i \in P} x_{ijkt})) \leq \sum_{i \in P} x_{ijkr} \quad \forall j \in P, \forall t, r \in T, t < r, \forall k \in N \\ & \sum_{i \in P} \sum_{j \in P} x_{ijkt} \leq M q_{pk} \quad \forall k \in N, \forall t \in T, \\ & x_{ijkt}, q_{kt}, d_{ipt}, cc_{itr} \in \{0, 1\} \quad \forall i, j \in P, \forall k \in N, \forall s \in S, \forall t \in T \\ & s_{ikt} \geq 0 \quad \forall i \in P, \forall k \in N, \forall t \in T \\ & e_{jt} \sum_{i \in P} x_{ijkt} \leq S_{ikt}(d, tt, st) \leq l_{jt} x_{ijkt} \quad \forall j \in P, \forall k \in N, \forall t \in T, \quad \forall d \in ext(\theta_d), \forall tt \in ext(\theta_{tt}), \forall st \in ext(\theta_{st}) \end{aligned}$$

$$\begin{aligned}
& S_{ikt}(d, tt, st) + d_{it}^k + tt_{ijt} + st_{it}^k - M(1 - x_{ijk}) \leq S_{jkt}(d, tt, st) \\
& \forall i, j \in P, \forall k \in N, \forall t \in T, \forall d \in \text{ext}(\vartheta_d), \forall tt \in \text{ext}(\vartheta_{tt}), \forall st \in \text{ext}(\vartheta_{st}) \\
& S_{ikt}(d, tt, st) - f_{ikt} - M(2 - av_{kt} + \sum_{i \in P} x_{ilk}) \leq M(1 - x_{djk(t+1)}) \\
& \forall j \in P, \forall k \in N, \forall t \in T, \forall d \in \text{ext}(\vartheta_d), \forall tt \in \text{ext}(\vartheta_{tt}), \forall st \in \text{ext}(\vartheta_{st})
\end{aligned}$$

4. Solving approach

In this section, we elaborate on our methodology for addressing the HHCSR with a Time window under uncertainty. The complexity of the problem at hand arises from its combinatorial nature. Furthermore, the analysis of the numerous constraints, both in terms of type and quantity, previously referenced in both deterministic and stochastic mathematical models, often in conflict, categorizes the tackled problem as NP-hard.

Moreover, to formally prove that the HHCSR is an NP-hard problem we can perform a reduction from the Vehicle Routing Problem (VRP) which is an NP-hard problem according to several works in the literature such as the paper of Bräysy and Gendreau (2005) that confirmed its solving in a polynomial time. The reduction is done following a step of steps starting by a conversion of each patient in the HHCSR problem to a city, then by mapping the caregivers to a depot city in the TSP problem. The travel distances must be set to be equal to the travel times between the locations of the patients in the scheduling problem. Based on the previous assumptions, if we can solve the VRP problem by visiting each city exactly once and returning to the central depot, this will present a valid schedule (solution) in the HHCSR problem where each patient is visited exactly once. Similarly, this is valid for the contrast sense, i.e., if we found a valid schedule in the HHCSR problem with exactly each patient is visited once, this will construct a VRP solution by following the corresponding route. To conclude, since the VRP is known to be NP-hard, that can be solved in a polynomial time, and we have demonstrated a reduction from the VRP to the HHCSR, we conclude that the HHCSR is also NP-hard.

In this case, it is obvious that the problem cannot be solved by relying on exact algorithms such as CPLEX and GUROBI software for handling large-size instances of the problem. Thus, to solve the studied problem efficiently, a three-level hybrid algorithm which will be executed sequentially. In the first step, we define initial feasible solutions that present a deterministic schedule using the greedy algorithm. The optimality and feasibility of the obtained solution will be examined in the next phase. These initial solutions will be rectified and improved using a VNS heuristic in the second phase to recover the feasibility and optimality.

VNS is well-suited for combinatorial optimization problems like scheduling, where the primary goal is to efficiently assign caregivers to patients while adhering to various constraints. It assists in generating high-quality schedules that respect time windows and other relevant constraints. The final phase adopts a Tabu Search (TS) metaheuristic as an enhancement mechanism, meticulously defining the optimal routes for nurses' home visits. The TS metaheuristic introduces an acceleration to convergence, refining each solution candidate through the implementation of three neighborhood search operations.

By utilizing VNS for scheduling and TS for routing, we harness the strengths of each metaheuristic in their respective domains. This division of labor enables us to achieve a more comprehensive and effective solution to the overall problem. Furthermore, this sequential approach aligns with real-world scenarios commonly encountered in healthcare logistics. In practice, healthcare providers often need to first establish a schedule for their caregivers based on patient appointments and then optimize the routes for caregivers to visit patients efficiently. This combined approach ensures that both the scheduling and routing components of the Home Health Care Scheduling and Routing Problem (HHCSR) are addressed optimally, resulting in a more robust and practical solution. The procedure of the proposed solution is illustrated in Fig. 2. The details of the three solving steps of the solution approach will be detailed in the next three subsections.

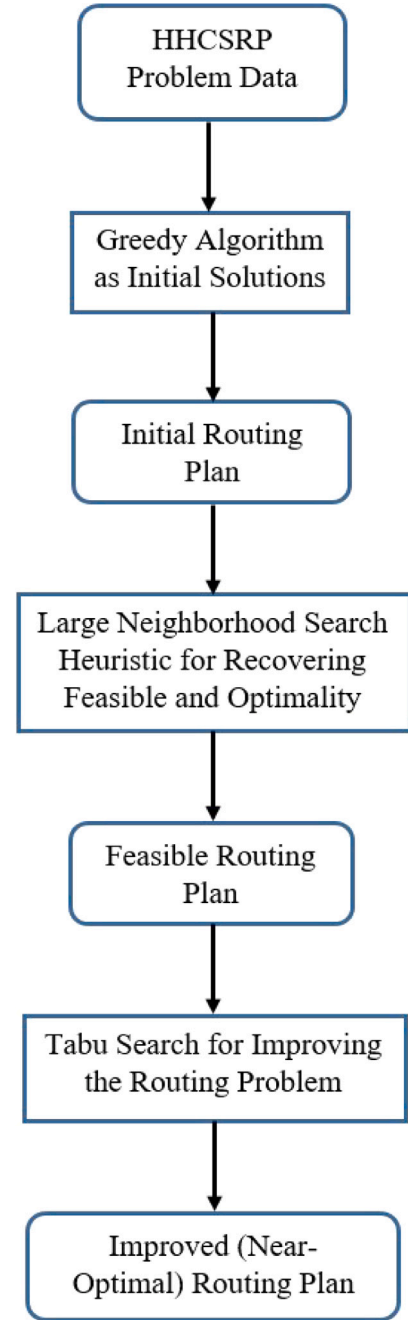


Fig. 2. The procedure of the proposed three-level hyperheuristic algorithm.

4.1. Initial solution using the Greedy algorithm

The initial solution is built by applying the greedy algorithm which assigns patients to nurses iteratively at their best condition until all patients are assigned. The choice of the greedy algorithm as an initialization procedure is explained by its simplicity and its low time complexity. Moreover, the greedy algorithm has been extensively in the literature in several fields and shows an ability to solve and find close optimal solutions for well-known optimization problems including the Travel Salesman Problem (Geng et al., 2011), the Huffman coding moreno2021lossless, the activity selection (Ranjan et al., 2021), the job sequencing (Zhao et al., 2021), etc. The Greedy method searches for the best choice at every phase and attempts to find the optimal way to handle the whole problem by searching for a locally optimal choice

with the aim of leading to a globally optimal solution in the future. In our context, the greedy algorithm will attempt to find a locally optimal choice for assigning the nurses to the patients aiming that this choice will get to a globally optimal solution. In the development of the initial solution, we only try to assign nurses to patients respecting the required skills and minimizing the total cost following the cost-least principle. The best assignment for each customer is decided by the feasibility in the resulting scheduling and the cost incurred to the existing schedules. However, when executing the greedy algorithm some of the patients will be not scheduled. This set of patients will be brought into their best position one after another until all of them will be reassigned. Thus, the initially generated solution may perform badly regarding other objectives such as minimizing the traveled distances, the workload balance, the nurses' consistency, and so on. However, we will improve the rest of the objectives using the VNS and the Tabu Search metaheuristics in the next phases. The overview of the greedy algorithm is presented in Algorithm 1.

Algorithm 1 Greedy Scheduling (Matrix [n] [m])

```

1. Input: Confusion Matrix of Nurse / Patients Assignment Skills
2. Output: Solution to the Assignment Problem
3. Begin
4.   Create a Matrix of Assignments[a][b] of Size [n][2]
5.   Initialize all the Items in Assignments[a][b] to 0.
   // We Make a Greedy Choice to Select an Item based on
   the Maximum Skill
6.   For (i = 1; i < n, i++)
7.     {
8.       maxSkill=1
9.       SelectedNurse=0
10.      For (j = 1; j < m, j++)
11.        {
12.          If Matrix [i][j]>maxSkill
13.            SelectedNurse=j
14.            maxSkill= Matrix [i][j]
15.          }
16.      Assignments[a][b]= Assignments[a][b] ∪ As-
signments[i][SelectedNurse]
17.    }
18.   Return Assignments[a][b]
19. End

```

4.2. Variable search neighborhood

The greedy heuristic is often considered naïve since it selects the easy insertions at the start of the process and makes complex assignments to the late iterations, which often lead to workload unbalance between caregivers. In addition, this may leads to a loss resulting from postponing an insertion to later iterations. Therefore, an adaptation phase is invoked to adjust the preschedule and improve the final solution. Thus, we opt to use a VNS heuristic which will take this loss into account. The main idea consists of deciding the assignment of each customer based on the maximum difference between his best assignment and other assignments. Moreover, each iteration in the algorithm is based on a selection and perturbation operators. The steps of the Variable Search Neighborhood heuristic are summarized in Algorithm 2.

Algorithm 2: Variable Search Neighborhood (Matrix [n] [m])

```

1. Input: Initial Solution S provided by the Greedy Scheduling
2. Output: An optimized Solution to the Assignment Problem
3. BEGIN
4.  $S_{best} = S$ 

```

```

5.   While Stopping Criteria not Reached do
6.      $S' = S$ 
7.     Use the Selection Operator to Select Assignments to
Adjust
8.     Use the Perturbation Operator to Remove Selected
Assignments
           from  $S'$  and Reinsert Unscheduled Visits as many
as Possible in  $S'$ .
9.     If  $f(S') > f(S_{best})$  then
10.       $S = S'$ 
11.       $S_{best} = S'$ 
12.     Else
13.       if  $f(s') > f(s)$  then
14.         $S = S'$ 
15.       Else
16.         $S = S'$  with a Probability  $p(S, S')$  ;
17.     End While
18. Return  $S_{best}$ 
19. END

```

4.3. Tabu search metaheuristic

In this third level of the solving approach, we select to implement a TS metaheuristic to define the optimal routes to be followed by caregivers to reach their customers' homes. Based on the initial routes generated in the previous step, the TS is then implemented in the second step to further minimize the traveling distances based on the obtained routes in the previous step. The TS was first introduced as an evolved optimization problem-solving by Glover (1989). It can be defined as a guided metaheuristic search process that accelerates convergence and optimizes each solution candidate through mutation operators and crossover operators. Moreover, The TS is based on penalizing moves that take the solution into "tabu" previously visited search space and accept non-improving solutions in order to prevent falling into local minimums.

We explain our choice by the following points:

- TS is known by its experimental performance when solving routing problems (Gendreau, 2012; Nanry & Barnes, 2000).
- The extreme performance when compared to similar metaheuristics, produces high-quality solutions with reasonable CPU times (Montané & Galvao, 2006; Rendl et al., 2012; Rest & Hirsch, 2016).
- The TS uses a reduced number of parameters (see Table 3).
- The algorithm can override local optimums by considering non-improving solutions.

As mentioned above (Section 4), Tabu Search uses three different neighborhood search operations to generate new candidate solutions that will be randomly performed. The neighborhood is determined by swapping a pair of customers within the same route called (i) Intra-route Insertion, or between two routes called (b) Inter-route Insertion, or (c) Inter-route Swap. These three neighborhood search operations are created using three kinds of move operators namely: exchange operator, 2-opt*, and 2opt operator.

- Intra-route Insertion:** used to generate new routes by moving a patient backward or forward or within a route. This neighborhood search operation chooses the patient randomly and then inserts it within the same route but into a different random location using the 2-opt move as can be depicted in Fig. 5.
- Inter-route Insertion:** This operation consists of randomly removing a patient from a routing plan and then inserting it into a different adjacent route. Each of the two selected routes is divided into two sections using the crossover to search the shortest paths. To do so, the algorithm uses a move function to decompose the set of selected routing plans into a set of sections.

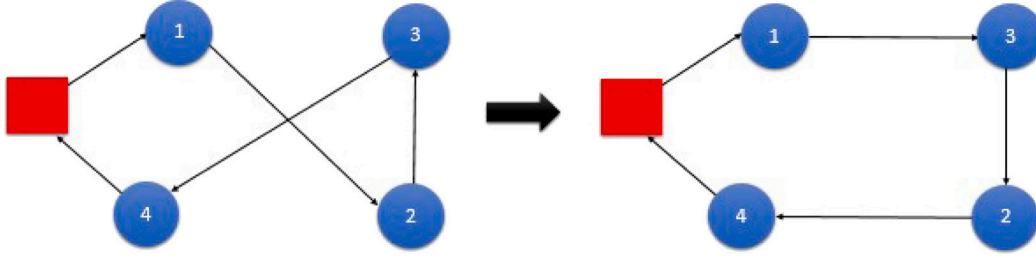


Fig. 3. Intra-route insertion using exchange operator.

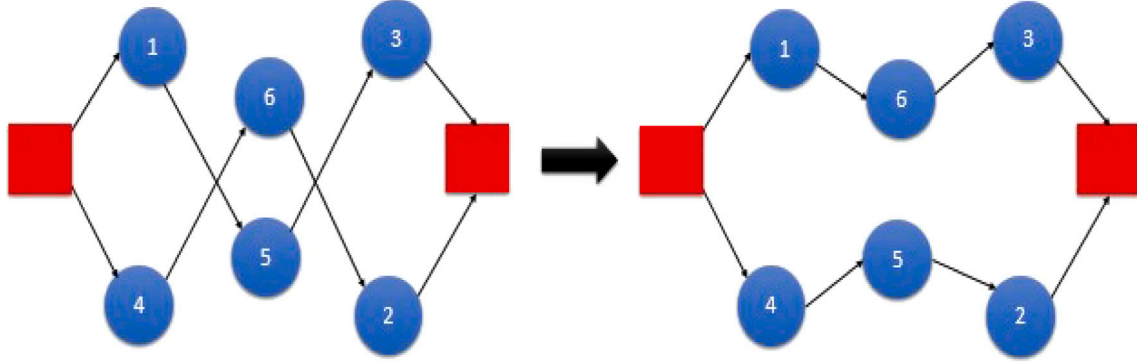


Fig. 4. Inter-route insertion using exchange operator.

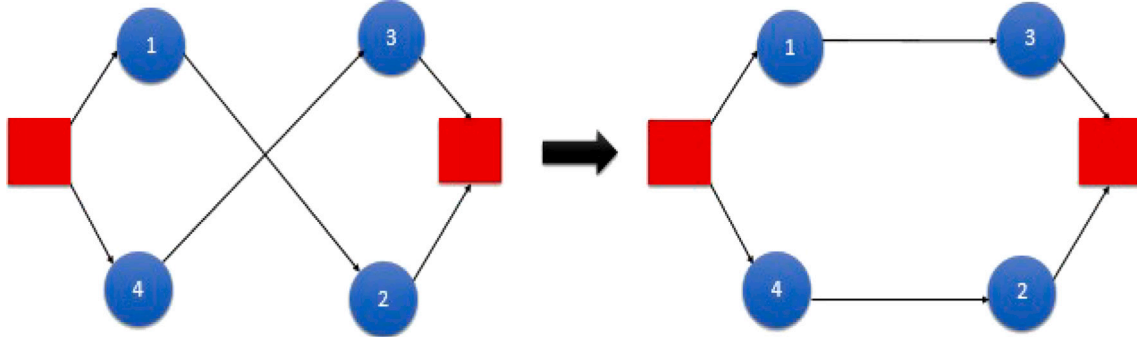


Fig. 5. Inter-route swap using 2-opt* operator.

After that, the patients are interchanged between two routes through the exchange move. By the end, a relocation movement removes a patient from one of the routes and inserts it in another route through the swap function. An illustrative example can be depicted in Fig. 4.

- (c) **Inter-route Swap:** This last operation will switch two patients between two different routes. The locations are precise randomly when the patients are inserted into the new scheduling. Moreover, the TS uses a 2-opt* function to permit moving from the current scheduling to another solution in its neighborhood, repeatedly as can be captured in Fig. 5. In this context, an anti-cycling technique should be invoked to investigate a broad solution space as the current solution may deteriorate during the search (see Fig. 3).

The pseudo-code of the proposed TS metaheuristic algorithm is provided in Algorithm 3 (Taşan Öz Mehmet, 2007).

Algorithm 3: Tabu Search Algorithm

1. **Input:** An Optimized Solution to the Assignment Component
2. **Output:** An Optimized Solution to the Routing Component

3. BEGIN

4. Step 1: Choose an Initial Solution i in S
5. Set $i^* = i$ and $k = 0$
6. Step 2: Set $k = k + 1$ and generate a subset V^* of solution in $N(i, k)$ such that either one of the Tabu conditions is violated or at least one of the inspiration conditions is satisfied.
7. Step 3: Choose a Best j in V^* and $i = j$
8. Step 4: **If** $f(i) < f(i^*)$ **Then**
9. set $i^* = i$
10. Step 5: Update Tabu and Inspiration Conditions
11. Step 6: **If** a Stopping Condition is Met **then**
12. Stop
13. **Else**
14. Go to Step 2.
15. **END**

In the following, we detail the parameters used by the TS algorithm:

5. Computational experiments and evaluation

In this section, we report the computational performance of the proposed modeling and solving approach that will be evaluated by

Table 3
Tabu search parameters and notations.

Notations	Description
i, j :	Indexes of the solution
k	Index of the iteration
V^*	Solution subset
$N(i, k)$	Neighborhood of the solution i at iteration k .
$f(i)$	Value of the objective function
Parameters	Value
Tabu list size	100
Stopping condition	1000
Operators neighborhood generation	2-opt Operator, 2-opt* Operator, and Swap Operator

conducting a series of experiments on Solomon's VRTPW benchmark (Solomon, 1987) which were adapted to our context and our model. We aim through the computational experiments to assess the performance of the suggested modeling and solving procedure by investigating the main characteristics outperforming similar procedures. In this context, our results will be compared to those obtained in Shahnejat et al. (2021) in which authors reported results of four approaches which are respectively, (i) CPLEX solution, (ii) the Simulated Annealing, (iii) Genetic Algorithm and (iv) a Memetic Algorithm (MA). Shahnejat et al. (2021)' solution is based on modeling a robust optimization model for an HHCSR with uncertain travel and service times. Both versions of the problem, i.e. deterministic and stochastic versions will be considered for solving and comparing.

To do that, we first detail the benchmark instances and the computational environment in Section 5.1. Then, Section 5.2 demonstrates the overall results concerning the total cost, the CPU running time, and the Number of used Vehicles (N.V). Finally, Section 5.3 analyzes the quality of the solutions regarding the best schedules using some statistical analysis.

5.1. Benchmark instances and computational environment

We will depict in this subsection the technical details of the three-level hybrid algorithm for solving the set of benchmark instances. Moreover, we will describe the set of data instances used to prove the efficiency of the suggested procedure (Solomon's VRTPW benchmarks). To do that, we will present in the next subsection the computational environment, while we will detail the benchmark instances used in the experiments in the second subsection will.

5.1.1. Computational environment

We coded the suggested algorithm in the Java programming language, i.e. the Greedy algorithm, the VNS, and the TS metaheuristic are all developed in Java. We used Eclipse 4.23 as an integrated development environment. All experiments were carried out on a Windows PC with a Core i5 – 2.50 GHz CPU and 8 GB RAM. For each benchmark, we have run the three-level hybrid algorithm 10 times, then the best solution is maintained.

5.1.2. Benchmark data instances

In order to build a consistent comparative study, we choose to use the same benchmarks as in Shahnejat et al. (2021). According to the authors, Solomon's VRTPW benchmark was adapted to fit with the suggested model. Based on the location distribution and the length of time windows, Solomon's benchmarks are classified into six categories respectively named R1, R2, C1, C2, RC1, and RC2. Each of the six groups is characterized according to specific features as follows:

- The R data instances are composed of the customer with randomly distributed locations.
- The C data instances contain customers with clustered locations.

- The RC data instances contain a mix of customers with random and clustered locations.
- The R1, C1, and RC1 data instances consider the narrow length of time windows.
- The R2, C2, RC2 data instances consider wider time windows.

Originally, Solomon's benchmark contains three sizes of 25, 50, and 100 (Solomon, 2023) customers and each data instance considers the service time, the demand, the time windows, and the location of each customer. To adapt the problem to our context, we considered the following assumptions:

- The vehicles are assumed as nurses and each one is associated with a skill S_i .
- The customers are assumed as patients and each patient demands a specified service characterized by the required skill.
- A location lab is added in each data instance with the coordinate (35, 45).
- The locations and the time windows are maintained. As cited in Shahnejat et al. (2021), the data instances in this study are named H-xxxx to separate them from Solomon's data instances.

5.2. Computational results

This subsection is devoted to reporting a set of computational experiments to prove the efficiency of our suggested modeling and solving approach compared to CPLEX software, SA, GA, and MA reported in Shahnejat et al. (2021). To do so, we will report results based on the model, i.e. (i) the experimental results for the deterministic model and (ii) the experimental results for the stochastic model. In both variants, the reported results are based on the computational efficiency in terms of three parameters which are:

- The Total Cost (TC): which reports and compares the objective function to be minimized by different approaches.
- The Computational Time (CT): which reports and compares the CPU time needed to reach a final solution to the problem by different approaches.
- The Number of Nurses: a pair that reports and compares the number of primary nurses (the first element of the pair), and the number of auxiliary nurses (the second element of the pair).

In addition, we added two gap columns to calculate the difference between our approach and compared approaches in terms of Total Cost (TC_Gap (%) column) and in terms of computational time needed to reach a final solution (CT_Gap (%) column). Both experimental results (for the deterministic and the stochastic models) will be reported in the following two subsections.

5.2.1. Experimental results of the deterministic model

To justify the efficiency of the suggested solution for solving the deterministic model of the problem using Solomon's benchmarks, we present the running test results in the following tables grouped by the size of the benchmark as follows:

- The Table 4 reports the first group of results by providing the running results of small-sized problem instances (25 patients) which can be solved using commercial software like CPLEX with an acceptable CPU running time. Moreover, results using SA, MA, and GA approaches to handle similar instances are reported, and their results are compared with our solution and optimal solutions achieved by CPLEX. As mentioned above, the key features reported in Table 4 are the total cost, the CPU running time, and the number of patients.

According to Table 4, the results obtained by our approach and the proposed three meta-heuristic algorithms in Shahnejat et al. (2021) are similar and very close to the exact technique solutions.

Table 4
Experimental results of the deterministic model for small-sized instances.

Instances	n	Specifications	Our approach			CPLEX					MA					GA					SA				
			TC	CT (s)	N.V	TC	CT (s)	N.V	TC_Gap (%)	CT_Gap (%)	TC	CT (s)	N.V	TC_Gap (%)	CT_Gap (%)	TC	CT (s)	N.V	TC_Gap (%)	CT_Gap (%)	TC	CT (s)	N.V	TC_Gap (%)	CT_Gap (%)
HHC-R101	4	(12,2,2)	308.54	4.09	(6,2)	308.54	50.13	(6,2)	0	1125.67	308.54	5.02	(6,2)	0	22.74	308.54	6.2	(6,2)	0	51.59	308.54	4.1	(6,2)	0	0.24
HHC-C101	4	(12,3,3)	319.16	4.12	(4,2)	319.16	44.77	(4,2)	0	986.65	319.16	6.8	(4,2)	0	65.05	319.16	6.38	(4,2)	0	54.85	319.16	4.2	(4,2)	0	1.94
HHC-RC101	4	(12,4,4)	290.47	5.9	(4,3)	290.47	51.15	(4,3)	0	766.95	290.47	6.16	(4,3)	0	4.41	290.47	7.31	(4,3)	0	23.90	290.47	6.16	(4,3)	0	4.41
HHC-R201	4	(15,5,2)	318.88	5.42	(3,2)	318.88	46.56	(3,2)	0	759.04	318.88	7.74	(3,2)	0	42.80	318.88	5.96	(3,2)	0	9.96	318.88	5.42	(3,2)	0	0.00
HHC-C201	4	(15,6,3)	255.61	5.03	(3,3)	255.61	51.36	(3,3)	0	921.07	255.61	5.02	(3,3)	0	−0.20	255.61	7.35	(3,3)	0	46.12	255.61	4.45	(3,3)	0	−11.53
HHC-RC201	4	(15,7,4)	313.97	4.91	(4,3)	313.97	40.61	(4,3)	0	727.09	313.97	6.38	(4,3)	0	29.94	313.97	6.41	(4,3)	0	30.55	313.97	5.02	(4,3)	0	2.24
HHC-R101	8	(20,7,2)	528.17	40.22	(6,3)	528.17	623.24	(6,3)	0	1449.58	528.17	52.53	(6,3)	0	30.61	538.2	43.92	(6,3)	1.9	9.20	543.43	42.98	(7,3)	2.89	6.86
HHC-C101	8	(20,8,3)	572.62	34.33	(5,4)	572.62	370.37	(5,4)	0	978.85	572.62	40.75	(5,4)	0	18.70	583.59	34.33	(5,4)	1.92	0.00	580.73	37.61	(5,4)	1.42	9.55
HHC-RC101	8	(20,9,4)	545.53	53.12	(4,5)	545.53	592.86	(4,5)	0	1016.08	545.53	53.16	(4,5)	0	0.08	545.53	49.02	(4,5)	0	−7.72	558.72	44.32	(4,5)	2.42	−16.57
HHC-R201	8	(22,9,2)	499.79	35.22	(3,3)	499.79	581.57	(3,3)	0	1551.25	499.79	40.53	(3,3)	0	15.08	512.48	36.77	(3,3)	2.54	4.40	507.14	43.21	(3,3)	1.47	22.69
HHC-C201	8	(22,10,3)	388.06	49.23	(4,4)	388.06	581.96	(4,4)	0	1082.12	388.06	62.8	(4,4)	0	27.56	388.06	55.34	(4,4)	0	12.41	392.86	51.88	(4,4)	1.24	5.38
HHC-RC201	8	(22,11,4)	529.19	43.99	(4,4)	529.19	577.35	(4,4)	0	1212.46	529.19	50.35	(4,4)	0	14.46	541.48	43.22	(4,4)	2.32	−1.75	529.19	44.54	(4,4)	0	1.25
Averages			23.7983			300.994			28.1033					25.1842					24.4908						

Table 5
Experimental results of the deterministic model for R1, C1, and RC1 large sized instances.

Instances	Specifications	Our approach			Memetic					Genetic					SA				
		TC	CT (s)	N.V	TC	CT (s)	N.V	TC_Gap (%)	CT_Gap (%)	TC	CT (s)	N.V	TC_Gap (%)	CT_Gap (%)	TC	CT (s)	N.V	TC_Gap (%)	CT_Gap (%)
H-R101	(150,10,40)	8949.91	428.49	(31,3)	8975.03	440.71	(31,3)	0.28	2.85	8991.29	406.07	(31,3)	0.46	-5.23	9219.29	392.71	(32,3)	3.01	-8.35
H-R102	(150,20,30)	9217.45	356.56	(26,6)	9287.55	368.67	(27,5)	0.76	3.40	9337.95	346.65	(26,6)	1.31	-2.78	9558.45	311.67	(28,5)	3.70	-12.59
H-R103	(150,30,20)	8387.05	455.82	(25,6)	8399.16	472.93	(25,6)	0.14	3.75	8546.73	435.28	(26,6)	1.90	-4.51	8762.73	429.93	(26,7)	4.48	-5.68
H-R104	(150,40,10)	8633.34	162.13	(16,9)	8642.04	164.24	(16,9)	0.10	1.30	8895.89	161.96	(17,9)	3.04	-0.10	9126.89	121.24	(16,9)	5.72	-25.22
H-R105	(200,10,90)	9236.06	542.95	(31,5)	9272.46	558.4	(31,5)	0.39	2.85	9410.96	533.75	(31,5)	1.89	-1.69	9664.96	510.4	(32,5)	4.64	-6.00
H-R106	(200,20,80)	11 623.87	491.27	(18,8)	11 704.32	484.06	(18,8)	0.69	-1.47	11 813.24	450.39	(18,8)	1.63	-8.32	12 091.24	414.06	(19,8)	4.02	-15.72
H-R107	(200,30,70)	11 139.74	561.46	(27,8)	11 099.52	549.94	(27,8)	-0.36	-2.05	11 430.74	532.23	(28,8)	2.61	-5.21	11 712.74	506.94	(27,9)	5.14	-9.71
H-R108	(200,40,60)	6759.65	389.26	(23,10)	6741.5	386.74	(23,10)	-0.27	-0.65	7039.2	371.01	(25,10)	4.14	-4.69	7297.2	321.74	(25,10)	7.95	-17.35
H-R109	(200,50,50)	7686.38	608.87	(22,11)	7746.9	594.55	(22,11)	0.79	-2.35	7795.32	570.29	(22,11)	1.42	-6.34	8103.32	549.55	(24,11)	5.42	-9.74
H-R110	(200,60,40)	9558.7	366.61	(23,11)	9513.48	352.09	(23,11)	-0.47	-3.96	9765.84	340.13	(24,11)	2.17	-7.22	10 027.84	287.09	(24,12)	4.91	-21.69
H-R111	(200,70,30)	9237.64	340.71	(23,9)	9302.64	353.06	(23,9)	0.70	3.62	9415.38	339.44	(23,9)	1.92	-0.37	9729.38	290.06	(24,10)	5.32	-14.87
H-R112	(200,80,20)	10 970.57	393.33	(19,10)	10 907.36	395.78	(19,10)	-0.58	0.62	11 189.26	387.71	(19,10)	1.99	-1.43	11 439.26	332.78	(21,10)	4.27	-15.39
H-C101	(150,10,40)	4216.91	547.04	(12,4)	4205.79	579.15	(12,4)	-0.26	5.87	4432.59	533.6	(14,4)	5.11	-2.46	4612.59	526.15	(15,4)	9.38	-3.82
H-C102	(150,20,30)	4280.5	598.81	(11,4)	4321.62	614.16	(11,4)	0.96	2.56	4370.9	606.54	(11,4)	2.11	1.29	4550.9	569.16	(12,4)	6.32	-4.95
H-C103	(150,30,20)	4005.12	271.13	(14,6)	4063.23	286.35	(14,6)	1.45	5.61	4142.48	274.51	(14,6)	3.43	1.25	4343.48	226.35	(15,7)	8.45	-16.52
H-C104	(150,40,10)	4805.18	268.83	(12,5)	4782.65	277.52	(12,5)	-0.47	3.23	5026.43	255.23	(13,5)	4.60	-5.06	5227.43	227.52	(14,5)	8.79	-15.37
H-C105	(200,30,70)	4738.45	597.19	(16,5)	4673.1	605.04	(16,5)	-1.38	1.31	4927.54	583.62	(17,5)	3.99	-2.27	5245.54	560.04	(18,5)	10.70	-6.22
H-C106	(200,40,60)	7391.82	193.64	(15,6)	7476.82	205.16	(15,6)	1.15	5.95	7675.24	200.89	(15,6)	3.83	3.74	7965.24	142.16	(16,6)	7.76	-26.59
H-C107	(200,50,50)	4743.35	336.21	(16,5)	4698.1	353.73	(16,5)	-0.95	5.21	5088.72	321.89	(18,5)	7.28	-4.26	5332.72	304.73	(18,5)	12.43	-9.36
H-C108	(200,60,40)	4786.25	226.71	(17,7)	4804.5	278.82	(17,7)	0.38	22.99	4827.84	275.13	(17,7)	0.87	21.36	5103.84	238.82	(18,7)	6.64	5.34
H-C109	(200,70,30)	6408.82	268.4	(13,10)	6462.82	253.2	(13,10)	0.84	-5.66	6577.88	244.59	(13,10)	2.64	-8.87	6885.88	189.2	(14,10)	7.44	-29.51
H-RC101	(150,10,40)	5103.58	255.25	(21,4)	5126.16	266.75	(21,4)	0.44	4.51	5158.52	248.65	(21,4)	1.08	-2.59	5358.02	225.75	(22,4)	4.99	-11.56
H-RC102	(150,20,30)	3566.88	490.07	(17,4)	3598.88	472.87	(17,4)	0.90	-3.51	3815.55	451.9	(18,4)	6.97	-7.79	4010.55	407.87	(19,4)	12.44	-16.77
H-RC103	(150,30,20)	3656.94	492.88	(15,5)	3698.94	487.65	(15,5)	1.15	-1.06	3887.94	451.85	(15,5)	6.32	-8.32	4091.94	420.65	(16,5)	11.90	-14.65
H-RC104	(150,40,10)	3438.84	425.73	(13,6)	3383.63	413.28	(13,6)	-1.61	-2.92	3643.43	400.96	(16,6)	5.95	-5.82	3877.43	367.28	(16,6)	12.75	-13.73
H-RC105	(200,30,70)	5466.09	661.62	(25,7)	5454.84	657.06	(25,7)	-0.21	-0.69	5773.84	604	(26,7)	5.63	-8.71	6015.84	587.06	(26,8)	10.06	-11.27
H-RC106	(200,40,60)	3929.17	614.37	(22,9)	4017.62	582.37	(22,9)	2.25	-5.21	4113.42	574.2	(22,9)	4.69	-6.54	4353.42	535.37	(23,9)	10.80	-12.86
H-RC107	(200,50,50)	2842.62	493.85	(20,8)	2902.8	482.65	(20,8)	2.12	-2.27	2944.36	440.97	(20,8)	3.58	-10.71	3188.36	416.65	(21,8)	12.16	-15.63
H-RC108	(200,60,40)	3916.9	811.2	(19,10)	3969.02	823.7	(19,10)	1.33	1.54	4201.54	764.57	(21,10)	7.27	-5.75	4493.54	774.7	(20,11)	14.72	-4.50
Total Avg.		6506.82	436.22	(18.72, 6.72)	6525.26	440.02	(19.41,6.9)	0.35	1.56	6697.93	417.52	(20.03,6.93)	3.44	-3.43	6944.48	385.78	(20.72,7.14)	7.80	-12.77

However, meta-heuristic algorithms need much less computational time than the CPLEX commercial solver. Moreover, our three-level hybrid approach needs a little less time than a similar metaheuristic approach, i.e. 23.80 s for our approach compared to 28.10 for the MA approach, 25.18 s for the GA approach, 24.49 s for SA approach which means a decrease of 8.19% of computational time needed to reach a final solution.

- **Tables 5 and 6** report the second group of results by providing the running results of large-sized problem instances (100 patients) which cannot be solved using CPLEX with an acceptable CPU running time. Thus, this set of experiments will report only metaheuristics approaches that present an excellent way to arrive at near-optimal solutions within a reasonable amount of time. Therefore, our three-level algorithm, SA, MA, and GA will be applied to 56 data instances and their results will be reported in **Tables 5 and 6**.

Based on the average of total cost and gaps provided in the two **Tables 5 and 6**, it can be depicted that the three-level hybrid approach outperform other approaches with a little more computational time. For instance, from **5**, it can be conducted that our approach results 6506.82 as an average total cost compared to 6525.26 for the MA approach, 6697.93 for the Genetic Algorithm, and 6944.48 for SA.

Similarly, it can be conducted **6** it can be conducted that our approach results less average total cost (7755.50) compared to 7767.07 for the MA approach, 7914.3 for the Genetic Algorithm, and 8152.55 for SA metaheuristic.

It should also be noted that our approach takes more time than its counterparts to reach the final solutions, i.e. 436.22 s for our algorithm compared to 417.5 for the Genetic metaheuristic and 385.8 for the SA metaheuristic. Moreover, we can see from **6** that our approach needs on average 435.58 s, in contrast, Genetic Algorithm needs 421.37 s and 367.32 s on average respectively to reach a final solution for the same data instances.

We can explain that by the fact that it contains three stages, which are considered longer and more complex than the rest of the approaches.

5.2.2. Experimental results of the stochastic model

In this subsection, the performance of the suggested three-level hybrid approach for the stochastic model is analyzed. To do so, we developed in the next subsections a set of running tests to investigate the impact of the length of time windows, the uncertainty of service time, the location distribution of patients, and the uncertainty of time windows on the achieved stochastic results. Then, the following indicators were defined to assess the efficiency of the obtained results;

- **V0** presents the probability (percentage) of satisfying all patients respecting the time window constraint.
- **V1** presents the probability of not satisfying at most one client respecting the time window constraint.
- **V2** presents the probability of not satisfying at most two clients respecting the time window constraint.
- **V3** presents the probability of not satisfying at most three patients respecting the time window constraint.

• Location Distribution Effect

To report the impact of location distribution on the obtained solution, we utilized the R1, C1, and RC1 instances, containing different types of patients' locations and considering stochastic demand, service, and travel times. The results of stochastic, deterministic and robust procedures for these benchmarks are reported in **Table 7**.

The results reported in **7** reveals a high efficiency of our proposed approach especially with V3 indicator that show that in many

data instances as with HR-110, HC-103, and HC-104, the probability of not serving a maximum of three patients given the time window constraint is 100%. Noteworthy, that our approach is also superior to the robust model proposed in [Shahnejat et al. \(2021\)](#), since it achieves 57.56%, 73.80, 88.39%, and 94.81% as a total average for V0, V1, V2, and V3 indicators respectively compared to 51.22%, 68.07%, 84.52%, and 92.82% for the model. However, this high degree of efficiency of our approach will cause an increase in the total cost objective function since our stochastic model results total 4266.10 as a total average with an increase of 7.69% compared to the robust model. Thus, the decision maker must pay attention to this matter in order to achieve balance and completeness between the total cost and the robustness of the obtained solution. It should be also noted here that the results indicate that the obtained deterministic solutions are very fragile since they report 1.06%, 4.02%, 5.63, and 6.96% as the total averages in solving R1, C1, RC1 datasets.

• Effect of the Time Windows Length

To report the impact of the time window length on serving patients, we utilized the R2, C2, and RC2 instances, containing different time window lengths. The results of stochastic, deterministic, and robust approaches for these instances are reported in **8**.

Similar to the results of the previous subsections, deterministic solutions still fragile when solving the data instances with a total average of 27.84% for V0 as an example. On the other hand, both robust and stochastic approaches show high efficiency in solving the data instances. For instance, both approaches succeed in obtaining 100% for all instances based on V0 indicator. Regarding the remaining three indicators, i.e. V0, V1, and V2, results indicate that they were are very close to 100%, as our approach reports 95.7%, 96.3%, 99.87% as a total average in solving R2, C2, and RC2 instances which slightly outperforms the robust approach proposed in [Shahnejat et al. \(2021\)](#) that realize respectively 93%, 97.1%, and 99%. However, this increase in the efficiency of our approach will generate an increase in the total cost parameter. In this context, our stochastic model results in a total of 4417.86 as a total average with an increase of 6.55% compared to the deterministic model and 3.03% against the robust model. This proves another time that, the decision maker must find a trade-off between the total cost and the robustness of the obtained solution.

In **Figs. 6, 7, 8, and 9**, we will simplify the results reported in **Tables 7 to 8** to investigate the efficiency of the developed stochastic procedure by comparing it to the deterministic model and the robust model. Accordingly, we will report graphically the value of the different indicators obtained through the three approaches for the large data set instances.

This set of figures indicates that the achieved results of the various indicators using the suggested stochastic model are almost superior to those obtained by the robust and deterministic models.

5.3. Statistical test analysis

This subsection is devoted to statistically assessing our proposed approach to solving the HHCSR against the deterministic and robust models. To do so, we will conduct a set of statistical tests analysis to show the high performance of the stochastic model compared to the deterministic and the robust models suggested in [Shahnejat et al. \(2021\)](#) by performing two statistical tests:

- To compare our results with other models, a Sign Test for pairwise comparisons will be used
- To justify that the significant performance in not happened randomly, a Wilcoxon signed-rank test will be used

All details will be provided in the next two subsections.

Table 6
Experimental results of the deterministic model for R2, C2, and RC2 large sized instances.

Instances	Specifications	Our approach			Memetic					Genetic					SA				
		TC	CT (s)	N.V	TC	CT (s)	N.V	TC_Gap (%)	CT_Gap (%)	TC	CT (s)	N.V	TC_Gap (%)	CT_Gap (%)	TC	CT (s)	N.V	TC_Gap (%)	CT_Gap (%)
H-R201	(150,10,40)	11 902	469.57	(13,3)	12 084.56	544.36	(13,3)	1.53	15.93	12 164.055	455.36	(13,3)	2.20	-3.03	12 410.055	410.36	(14,3)	4.27	-12.61
H-R202	(150,20,30)	9928.04	258.19	(11,3)	10 208.49	325.98	(11,3)	2.82	26.26	10 310.25	243.98	(11,3)	3.85	-5.50	10 551.75	189.98	(12,3)	6.28	-26.42
H-R203	(150,30,20)	11 216.27	281.03	(11,4)	11 468.79	308.82	(11,4)	2.25	9.89	11 480.715	266.82	(11,4)	2.36	-5.06	11 711.715	211.82	(12,4)	4.42	-24.63
H-R204	(150,40,10)	10 049.31	230.2	(5,4)	10 274.46	303.99	(5,4)	2.24	32.05	10 509.045	215.99	(6,4)	4.57	-6.17	10 710.05	163.99	(6,4)	6.57	-28.76
H-R205	(200,20,80)	12 540.36	446.82	(12,3)	12 695.9	492.61	(12,3)	1.24	10.25	12 841.02	432.61	(12,3)	2.40	-3.18	13 057.02	379.61	(13,3)	4.12	-15.04
H-R206	(200,30,70)	14 702.76	377.65	(10,3)	14 882.88	446.44	(10,3)	1.23	18.22	14 944.36	363.44	(10,3)	1.64	-3.76	15 206.36	308.44	(11,3)	3.43	-18.33
H-R207	(200,40,60)	13 607.42	358.46	(8,3)	13 797.54	420.25	(8,3)	1.40	17.24	13 823.6	344.25	(8,3)	1.59	-3.96	14 043.6	289.25	(8,3)	3.21	-19.31
H-R208	(200,50,50)	7 645.23	362.53	(9,4)	7 657.78	418.32	(9,4)	0.16	15.39	7 853.1	348.32	(10,4)	2.72	-3.92	8 091.1	285.32	(11,4)	5.83	-21.30
H-R209	(200,60,40)	8 797.42	435.76	(8,4)	9 057.94	471.55	(8,4)	2.96	8.21	9 243.7	421.55	(9,4)	5.07	-3.26	9 495.7	360.55	(9,4)	7.94	-17.26
H-R210	(200,70,30)	10 245.07	412.36	(7,4)	10 439.32	451.15	(7,4)	1.90	9.41	10 446.06	398.15	(7,4)	1.96	-3.45	10 716.8	347.15	(8,4)	4.60	-15.81
H-R211	(200,80,20)	11 813.17	387.97	(10,4)	12 036.12	440.76	(10,4)	1.89	13.61	12 354.04	373.76	(11,4)	4.58	-3.66	12 568.04	312.76	(11,4)	6.39	-19.39
H-C201	(150,10,40)	5 178.86	374.03	(7,2)	5 449.01	422.82	(7,2)	5.22	13.04	5 511.72	359.82	(7,2)	6.43	-3.80	5 700.72	292.82	(8,2)	10.08	-21.71
H-C202	(150,20,30)	4 180.31	401.45	(8,3)	4 375.83	458.09	(8,3)	4.68	14.11	4 397.085	387.24	(8,3)	5.19	-3.54	4 652.09	346.8	(9,3)	11.29	-13.61
H-C203	(150,30,20)	4 102.66	377.29	(9,3)	4 299.78	419.08	(9,3)	4.80	11.08	4 347.84	363.08	(9,3)	5.98	-3.77	4 554.84	309.08	(10,3)	11.02	-18.08
H-C204	(150,40,10)	6 397.8	332.73	(8,3)	6 327.62	398.52	(8,3)	-1.10	19.77	6 496.335	318.52	(9,3)	1.54	-4.27	6 707.84	248.52	(10,3)	4.85	-25.31
H-C205	(200,30,70)	5 617.91	522.06	(11,3)	5 474.24	571.85	(11,3)	-2.56	9.54	5 530.32	507.85	(11,3)	-1.56	-2.72	5 868.32	460.85	(13,3)	4.46	-11.72
H-C206	(200,40,60)	9 932.87	504.4	(10,3)	9 631.02	580.19	(10,3)	-3.04	15.03	9 907.2	490.19	(11,3)	-0.26	-2.82	10 173.2	439.19	(11,3)	2.42	-12.93
H-C207	(200,50,50)	5 159.22	415.05	(8,3)	4 964.16	459.84	(8,3)	-3.78	10.79	5 104.44	400.84	(10,3)	-1.06	-3.42	5 392.44	343.84	(11,3)	4.52	-17.16
H-C208	(200,60,40)	6 430.28	413.25	(10,4)	6 192.73	442.04	(10,4)	-3.69	6.97	6 362.76	399.04	(11,4)	-1.05	-3.44	6 578.76	348.04	(11,4)	2.31	-15.78
H-RC201	(150,10,40)	5 545.53	569.13	(11,2)	5 312.04	627.92	(11,2)	-4.21	10.33	5 521.38	554.92	(12,2)	-0.44	-2.50	5 728.38	503.92	(13,2)	3.30	-11.46
H-RC202	(150,20,30)	4 509.225	530.33	(12,3)	4 233.45	562.12	(12,3)	-6.12	5.99	4 464.975	516.12	(13,3)	-0.98	-2.68	4 680.98	455.12	(14,3)	3.81	-14.18
H-RC203	(150,30,20)	4 157.81	351.26	(7,3)	4 232.81	403.05	(7,3)	1.80	14.74	4 285.425	337.05	(7,3)	3.07	-4.05	4 444.43	291.05	(8,3)	6.89	-17.14
H-RC204	(150,40,10)	4 258.78	369.57	(7,3)	4 013.3	396.36	(7,3)	-5.76	7.25	4 170.66	355.36	(8,3)	-2.07	-3.85	4 380.66	299.36	(9,3)	2.86	-19.00
H-RC205	(200,30,70)	7 273.38	505.92	(15,3)	7 024.32	550.71	(15,3)	-3.42	8.85	7 251.24	491.71	(16,3)	-0.30	-2.81	7 509.24	433.71	(16,3)	3.24	-14.27
H-RC206	(200,40,60)	5 344.79	550.6	(16,4)	5 175.64	603.39	(16,4)	-3.16	9.59	5 526.22	536.39	(18,4)	3.39	-2.58	5 800.22	490.39	(19,4)	8.52	-10.94
H-RC207	(200,50,50)	3 338.42	779.77	(14,4)	3 040.84	833.56	(14,4)	-8.91	6.90	3 323.96	765.56	(16,4)	-0.43	-1.82	3 541.96	723.56	(16,4)	6.10	-7.21
H-RC208	(200,60,40)	5 523.52	743.33	(13,4)	5 360.28	780.12	(13,4)	-2.96	4.95	5 514.7	729.12	(13,4)	-0.16	-1.91	5 842.4	765.12	(14,4)	5.77	2.93
Total Avg.		7755.50	435.58	(10,3.3)	7767.07	486.44	(10, 3.30)	0.15	11.68	7914.3	421.37	(10.63, 3.30)			8152.55	367.32	(11.37, 3.30)		

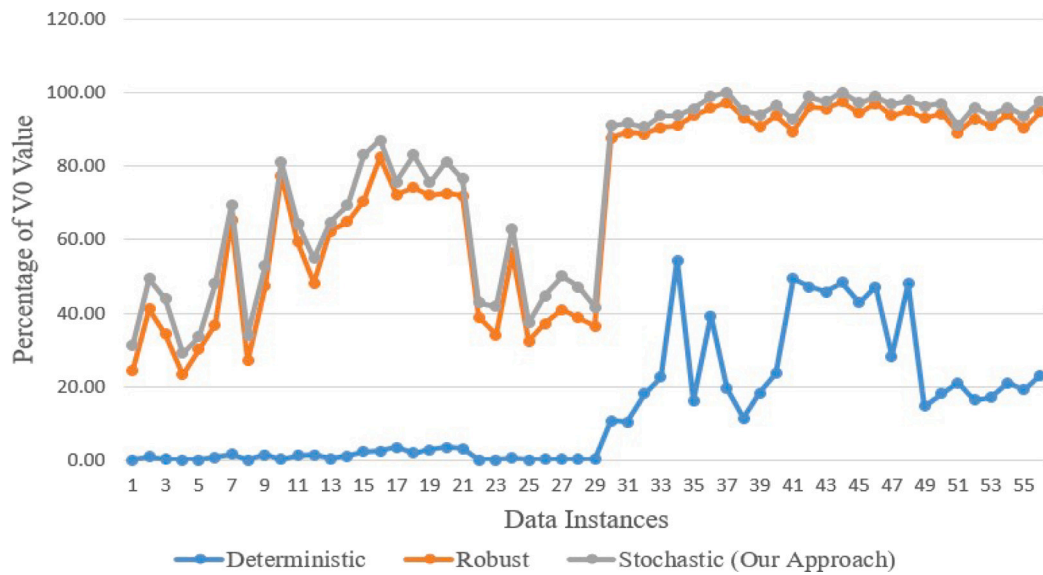


Fig. 6. Value of V0 achieved by the Stochastic, Robust, and Deterministic models for large data instances.

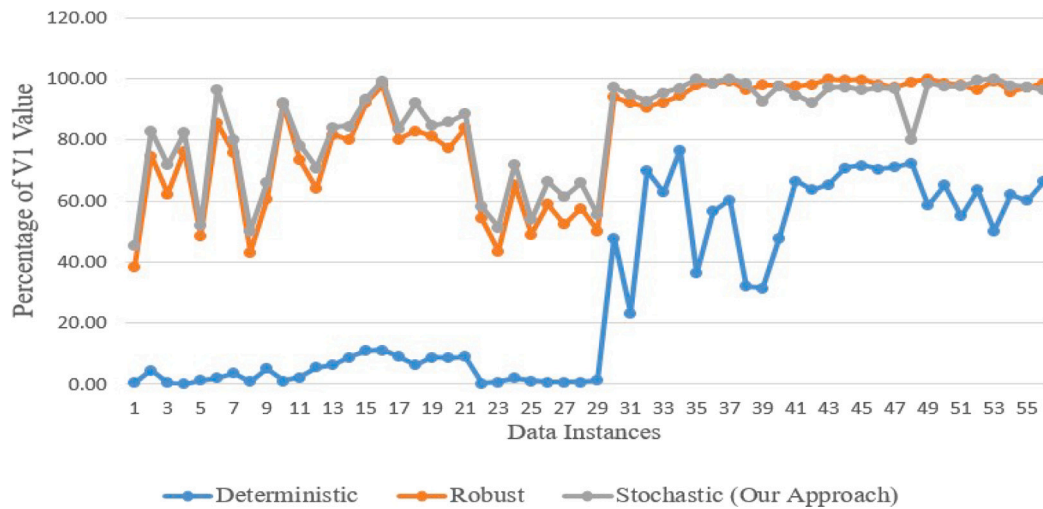


Fig. 7. Value of V1 achieved by the Stochastic, Robust, and Deterministic models for large data instances.

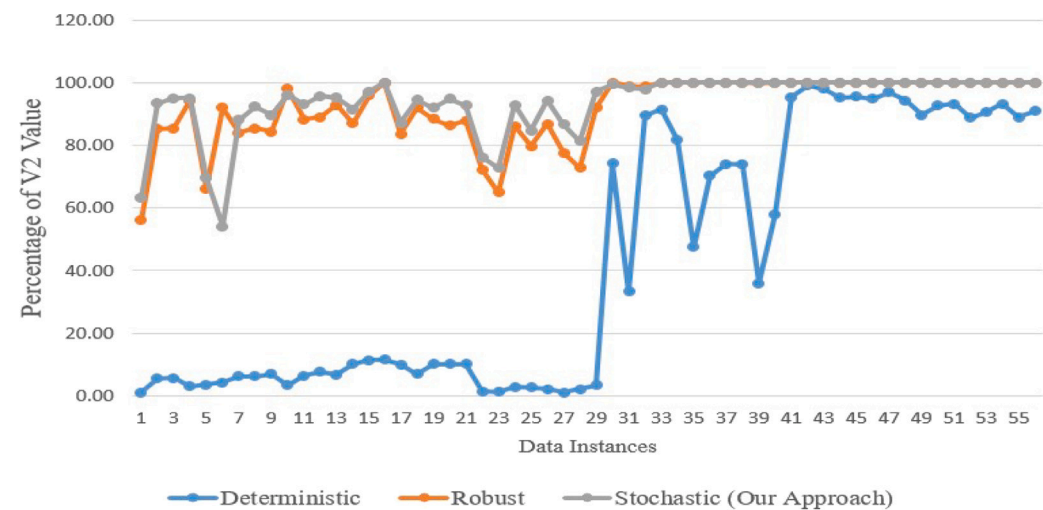


Fig. 8. Value of V2 achieved by the Stochastic, Robust, and Deterministic models for large data instances.

Table 7

Experimental results of the stochastic model for R1, C1, and RC1 large sized instances.

Instance	Our approach					Deterministic model					Increase (%)	Robust model					Increase (%)
	TC	V0	V1	V2	V3	TC	V0	V1	V2	V3		TC	V0	V1	V2	V3	
H-R101	7284.07	31.37	45.40	63.13	95.99	5983.35	0.05	0.49	1.15	3.79	21.74	6963.95	24.37	38.40	56.13	88.99	4.60
H-R102	7322.58	49.38	82.88	93.38	99.22	6191.70	0.94	4.55	8.23	18.26		7032.06	41.18	74.68	85.18	91.02	4.13
H-R103	6751.5	43.95	71.88	95.1	84.12	5599.44	0.14	0.52	5.71	7.61	20.57	6490.75	34.25	62.18	85.40	98.15	4.02
H-R104	6981.24	29.08	82.28	95.12	95	5761.36	0.04	0.17	3.13	5.95	21.17	6769.12	23.18	76.38	94.26	100.00	3.13
H-R105	4750.82	33.54	52.02	69.61	95.05	4636.23	0.10	1.21	3.58	7.28	2.47	4730.70	30.14	48.62	66.21	91.65	0.43
H-R106	6844.18	47.94	96.51	54.12	92.1	5852.16	0.64	1.99	4.24	6.91	16.95	6626.06	36.84	85.41	92.00	97.14	3.29
H-R107	5782.76	69.32	79.98	88.17	98.8	5549.76	1.57	3.71	6.19	8.00	4.20	5684.64	65.12	75.78	83.97	94.60	1.73
H-R108	3745.47	34.11	50.1	92.41	99	3370.75	0.04	0.87	6.24	7.77	11.12	3621.35	27.11	43.10	85.41	92.00	3.43
H-R109	4291.8	52.75	65.95	89.69	95.19	3873.45	1.49	5.19	7.00	7.67	10.80	4145.95	47.35	60.55	84.29	89.79	3.52
H-R110	5550.72	80.92	92.24	96.1	100	4756.74	0.12	1.13	3.37	6.08	16.69	5310.60	77.32	91.62	98.00	100.00	4.52
H-R111	4944.1	64.27	78.32	93.11	96.17	4651.32	1.24	2.24	6.51	8.05	6.29	4831.98	59.37	73.42	88.21	91.27	2.32
H-R112	5891.64	54.82	70.79	95.7	91.52	5453.68	1.42	5.51	7.60	8.05	8.03	5691.52	48.12	64.09	89.00	94.00	3.52
Avg.	5845.07	49.2875	72.3625	85.47	95.18		0.65	2.30	5.02	7.12		5658.22	42.86	66.19	84.01	94.05	
H-C101	3507.83	64.75	84.08	95.25	95.52	2803.86	0.45	6.35	6.79	7.75	25.11	3187.71	62.35	81.68	92.85	97.63	10.04
H-C102	4032.53	69.36	84.5	91.5	99.33	2881.08	1.12	8.94	10.26	10.46	39.97	3572.41	64.86	80.00	87.00	94.83	12.88
H-C103	4540.05	83.1	93.51	97.2	100	2708.82	2.34	11.09	11.47	12.11	67.60	3919.93	70.60	92.00	96.00	99.80	15.82
H-C104	4172.27	86.94	99.2	100	100	3188.43	2.50	11.22	11.60	12.70	30.86	3772.15	82.44	98.00	100.00	100.00	10.61
H-C105	2925.06	75.73	83.71	87.22	92.33	2336.55	3.52	9.06	9.92	11.21	25.19	2684.54	72.23	80.21	83.72	88.83	8.96
H-C106	6339.97	83.06	92	94.5	95.8	3738.41	1.92	6.30	6.95	9.60	69.59	5589.85	74.06	83.00	92.00	97.40	13.42
H-C107	3023.89	75.54	84.7	91.98	97.73	2349.05	2.72	8.87	10.20	11.57	28.73	2707.77	72.04	81.20	88.48	94.23	11.67
H-C108	3046.04	81.1	85.78	94.88	95.4	2402.25	3.49	8.74	10.15	11.72	26.80	2740.79	72.60	77.28	86.38	98.97	11.14
H-C109	4002.45	76.51	88.54	92.69	96.45	3231.41	3.17	9.05	10.19	11.82	23.86	3657.89	71.81	83.84	87.99	99.48	9.42
Avg.	3954.45	77.34	88.45	93.91	96.95		2.36	8.85	9.72	10.99		3537.00	71.44	84.13	90.49	96.80	
H-RC101	4766.17	42.89	58.26	76.2	86.61	3417.44	0.05	0.25	1.45	2.17	39.47	4156.05	38.89	54.26	72.20	82.61	14.68
H-RC102	3513.13	41.8	51.39	72.8	81.82	2399.25	0.03	0.63	1.25	1.87	46.43	3009.01	34.00	43.59	65.00	74.02	16.75
H-RC103	3206.99	63	71.93	92.98	99.3	2465.96	0.51	2.15	2.82	3.48	30.05	2864.21	56.20	65.13	86.18	92.50	11.97
H-RC104	2535.87	37.32	53.89	84.71	86.72	2255.75	0.08	1.14	2.72	3.07	12.42	2431.61	32.42	48.99	79.81	81.82	4.29
H-RC105	3406.93	44.7	66.38	94.38	92.74	2727.42	0.13	0.74	2.22	3.49	24.91	3092.52	37.20	58.88	86.88	94.18	10.17
H-RC106	2171.78	49.99	61.38	86.62	98.56	2008.81	0.16	0.62	1.13	1.62	8.11	2067.24	40.99	52.38	77.62	89.56	5.06
H-RC107	1987.29	47.16	66.04	81.37	97.35	1451.40	0.12	0.62	2.16	2.94	36.92	1667.17	38.66	57.54	72.87	88.85	19.20
H-RC108	2402.22	41.62	55.43	97.18	95.4	1984.51	0.24	1.22	3.35	3.59	21.05	2226.84	36.42	50.23	91.98	97.38	7.88
Avg.	2998.80	46.06	60.59	85.78	92.3125		0.16	0.92	2.14	2.78		2689.33	39.35	53.88	79.07	87.62	
Total Avg.	4266.11	57.56	73.80	88.39	94.81		1.06	4.02	5.63	6.96		3961.52	51.22	68.07	84.52	92.82	

Table 8

Experimental results of the stochastic model for R2, C2, and RC2 large sized instances.

Instance	Our approach					Deterministic model					Increase (%)	Robust model					Increase (%)
	TC	V0	V1	V2	V3	TC	V0	V1	V2	V3		TC	V0	V1	V2	V3	
R201	8450.64	90.98	97.18	99.5	100	8056.37	10.76	47.91	74.19	86.51	4.89	8320.53	87.78	93.98	99.98	100	1.56
R202	7052.07	91.73	94.81	98.4	100	6805.66	10.32	23.25	33.31	45.8	3.62	6921.96	89.03	92.11	99	100	1.88
R203	7788.88	90.63	92.62	97.96	100	7645.86	18.15	69.77	89.6	94.01	1.87	7658.77	88.73	90.72	98.69	99.99	1.70
R204	7125.91	93.72	95.4	100	100	6849.64	22.63	62.8	91.4	97.76	4.03	6995.8	90.52	92.2	100	100	1.86
R205	6657.23	93.65	97.02	100	100	6347.95	54.44	76.73	81.63	93.89	4.87	6527.12	90.95	94.32	100	100	1.99
R206	7829.79	95.59	99.81	100	100	7441.44	16.12	36.28	47.55	59.86	5.22	7699.68	93.69	97.91	100	100	1.69
R207	7486.52	98.95	98.57	100	100	6898.77	39.18	56.67	70.39	82.85	8.52	7356.41	95.75	98.33	100	100	1.77
R208	4159.18	100	100	100	100	3828.89	19.46	60.33	74.08	78.62	8.63	4029.07	97.3	99.33	100	100	3.23
R209	5049.39	95.18	98.27	100	100	4528.97	11.36	31.97	73.92	86.29	11.49	4919.28	93.28	96.37	100	100	2.64
R210	5471.89	93.91	92.75	100	100	5219.66	18.21	31.55	35.89	45.37	4.83	5341.78	90.71	97.94	99.93	100	2.44
R211	6278.45	96.47	97.64	100	100	6018.06	23.59	47.82	57.95	70.45	4.33	6148.34	93.77	97.64	100	100	2.12
Avg.	6668.18	94.62	96.73	99.62	100	6331.02	22.2	49.55	66.36	76.49		6538.07	95.53	100	99.78	100	1.99
C201	3820.98	92.66	94.57	100	100	3632.67	49.42	66.54	95.33	98.99	5.18	3690.87	89.46	97.59	100	100	3.53
C202	3165.91	98.81	92.17	100	100	2917.22	47.15	63.62	99.18	100	8.52	3035.8	96.11	98.03	100	100	4.29
C203	3054.2	97.57	97.11	100	100	2866.52	45.75	65.29	98.09	100	6.55	2924.09	95.67	99.79	100	100	4.45
C204	4467.84	100	97.41	100	100	4218.41	48.42	70.82	95.2	99.87	5.91	4337.73	97.59	99.42	100	100	3.00
C205	2959.98	97.09	96.45	100	100	2723.5	42.89	71.59	95.49	98.47	8.68	2829.87	94.39	99.68	100	100	4.60
C206	5044.56	98.89	97.11	100	100	4791.55	47.06	70.4	94.86	98.08	5.28	4914.45	96.99	98.15	100	100	2.65
C207	2754.25	96.86	96.75	100	100	2469.73	28.28	71.01	97.05	99.31	11.52	2624.14	93.66	97.05	100	100	4.96
C208	3302.53	97.8	80.11	100	100	3080.96	47.98	72.4	94.38	99.74	7.19	3172.42	95.1	98.95	100	100	4.10
Avg.	3571.28125	97.46	93.96	100	100	3337.57	44.62	68.96	96.2	99.31		3441.17125	94.87	98.58	100	100	
RC201	3848.54	96.23	98.54	100	100	3541.36	14.58	58.48	89.62	96.62	8.67	3718.43	93.03	99.81	100	100	3.50
RC202	3065.06	97	97.54	100	100	2822.3	18.1	65.14	92.74	97.7	8.60	2934.95	94.3	98.43	100	100	4.43
RC203	3065.28	91.03	97.45	100	100	2821.87	21.03	54.98	93.19	97.7	8.63	2935.17	89.13	97.93	100	100	4.43
RC204	2939.35	96.04	99.52	100	100	2675.53	16.38	63.61	88.83	97.34	9.86	2809.24	92.84	96.32	100	100	4.63
RC205	3665.55	93.61	100	100	100	3512.16	17.11	50.15	90.56	96.72	4.37	3535.44	90.91	99.35	100	100	3.68
RC206	2847.32	96.02	97.65	100	100	2587.82	20.98	62.12	93.32	97.7	10.03	2717.21	94.12	95.75	100	100	4.79
RC207	1728.04	93.54	97.42	100	100	1520.42	19.17	60.17	88.99	97.68	13.66	1597.93	90.34	97.16	100	100	8.14
RC208	2954.01	97.5	96.4	100	100	2680.14	23.12	66.44	91	97.7	10.22	2823.9	94.8	98.47	100	100	4.61
Avg.	3014.14	95.12	98.07	100.00	100	2770.2	18.81	60.14	91.03	97.4		2884.03	92.43	97.9	100	100	
Total Avg.	4417.87	95.73	96.25	99.87	100	4146.26	27.84	58.44	82.51	89.45		4287.76	92.96	97.14	99.91	100	

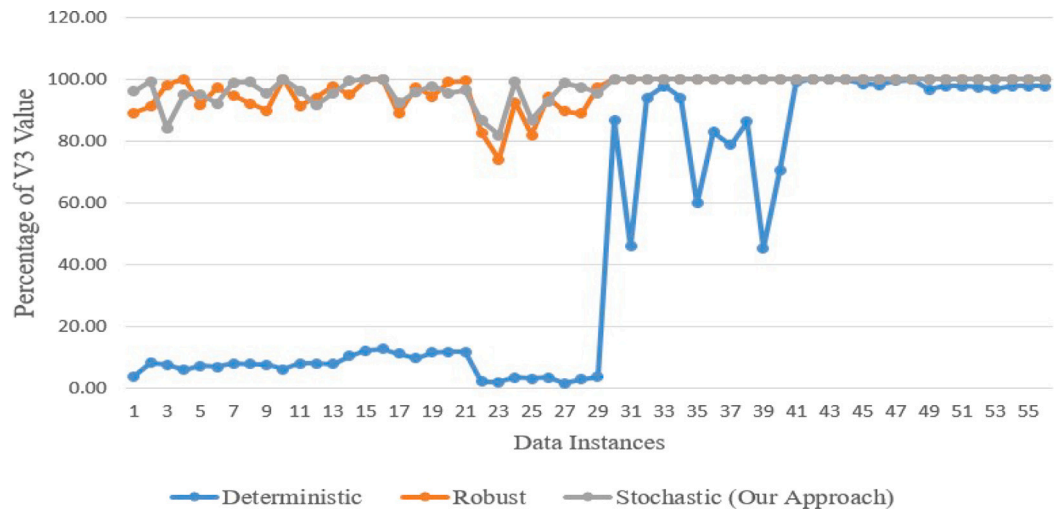


Fig. 9. Value of V3 achieved by the Stochastic, Robust, and Deterministic models for large data instances.

Table 9
Pairwise comparisons using Sign Test for the deterministic and the stochastic models based on the computational time and the total cost metrics.

Metric	Model	Data instances	Computational time comparison				
			Comparisons	CPLEX	MA	GA	SA
Computational time	Deterministic	Small sized	+	12	11	9	9
			–	0	1	2	2
			Equal	0	0	1	1
		R1, C1, RC1	+	NA	17	4	0
			–	NA	12	25	29
			Equal	NA	0	0	0
		R2, C2, RC2	+	NA	27	0	1
			–	NA	0	27	26
			Equal	NA	0	0	0
	Stochastic	Data instances	Comparisons	Column-and-row generation	MA		
		Small sized	+	12	9		
			–	0	3		
			Equal	0	0		
Total cost	Deterministic	Data instances	Comparisons	CPLEX	MA	GA	SA
		Small sized	+	0	0	4	5
			–	0	0	0	0
			Equal	12	12	8	8
		R1, C1, RC1	+	NA	19	27	27
			–	NA	10	0	0
			Equal	NA	0	0	0
		R2, C2, RC2	+	NA	15	17	27
			–	NA	12	10	0
			Equal	NA	0	0	0
	Stochastic	Data instances	Comparisons	Deterministic model	Robust model		
		R1, C1, RC1	+	0	0		
			–	29	29		
			Equal	0	0		
		R2, C2, RC2	+	0	0		
			–	27	27		
			Equal	0	0		
		Data instances	Comparisons	Deterministic model	Robust model		
		Small sized	+	0-	0		
			–	0	0		
			Equal	12	12		

NA: means Not Applicable, i.e., the CPLEX is enable to solve the considered data instances since it is large.

Table 10

Experimental results for R1, C1, and RC1 data instances containing 100 patients.

Indicator values	Deterministic model				Robust model			
	V0	V1	V2	V3	V0	V1	V2	V3
+	29	29	29	29	29	29	27	19
−	0	0	0	0	0	0	2	10
Equal	0	0	0	0	0	0	0	0

Table 11

Experimental results for R2, C2, and RC2 data instances containing 100 patients.

Indicator values	Deterministic model				Robust model			
	V0	V1	V2	V3	V0	V1	V2	V3
+	29	29	29	27	29	29	27	19
−	0	0	0	0	0	0	2	10
Equal	0	0	0	2	0	0	0	0

which requires the attention of the decision-maker in order to achieve balance and completeness between the total cost and the robustness of the obtained solution.

In addition to the previous two metrics, we will incorporate a third comparison metric based on indicator values to not only enriches our assessment but also provides a more holistic evaluation of our approach in relation to similar methodologies.

- **Indicator Values:** These depend on measuring the percentage of serving all patients, the percentage of not serving at most one patient, the probability of not serving at most two patients, and the percentage of not serving at most three patients respecting the time window constraints. These are important indicators as serving the largest possible number of patients is a major goal because unserved patients in results penalties. This metric is reported in [Tables 10, 11](#) shows the pairwise comparisons between our model, the deterministic model, and the robust model. The results provided in [Tables 10, 11](#), prove the superiority of our approach against the deterministic and the robust models similar approaches solving deterministic model with large models datasets. It is important to mention here that our approach fails to give better results in 10 out of 29 cases against the robust model and in 2 out of 29 against the deterministic model.

5.3.2. Wilcoxon Signed-Rank test

This part is devoted to inferring the real significance of the obtained results based on the deterministic, robust and stochastic models. However, a direct comparison of results may tell us little about the significant differences and their importance. A good way to determine the significance of the results is to develop a statistical significance test. The statistical significance test that we use to analyze our results is the non-parametric statistical test Wilcoxon signed-rank that is highly recommended to assess whether the obtained results of our suggested method are significant or not, i.e. it states if the obtained results are random or not, in terms of supporting the investigated idea. The utilization of the Wilcoxon Signed-Rank Test will leverage the outcomes obtained from the Sign Test for Pairwise Comparisons, as reported in [Section 5.3.1](#), to demonstrate the statistical significance of the observed results.

In our paper, the test hypothesis are:

Null Hypothesis H0: The probability of satisfying patients (V0, V1, V2, V3) of our approach is equal to the probability of satisfying patients by compared approaches.

Alternative Hypothesis H1: The probability of satisfying patients (V0, V1, V2, V3) of our procedure is different from the probability of satisfying patients by compared approaches.

We then report the different obtained p-values after executing the Wilcoxon signed-rank test on the R1, C1, RC1, R2, C2, RC2 in [Tables from 12 to 17](#). Each table reports the related statistical results (*p*-value, Standard error, Test statistic, etc.) achieved by the stochastic model compared to the deterministic and robust models for each Solomon's benchmark independently.

The rejection of the Null Hypothesis depends on the *p*-value, i.e. the results are not happened by random chance if the *p*-value is lower than 0.05 which means strong evidence against the null hypothesis, and by result the alternative hypothesis must be accepted which confirms that our results are statistically significant.

According to results provided in [Tables 12 to 17](#) reporting the different test statistics and p-values obtained when conducting the Wilcoxon statistical test on R1, C1, RC1, R2, C2, and RC2 data instances, the majority of p-values are less than 0.05 (32 cases out of 35 cases), indicating that our obtained solution definitely shows better and significant performance over the deterministic and the robust models.

6. Conclusion

In this study, we have investigated an integrated scheduling and routing problem under uncertainty, a problem highly arising in the home healthcare sector taking into account stochastic demand, service, and travel times. We build a two-stage mixed integer mathematical formulation to model the HHCSR. We also suggested a TS metaheuristic embedding a VNS heuristic to solve the model. The efficiency and effectiveness of the modeling and solving approach are proven through computational experiments on small and large-sized problem instances. Finally, a comparison study and statistical tests are performed to measure the performance of the solution against results obtained by CPLEX commercial solver and similar approaches used in [Shahnejat et al. \(2021\)](#). The results show that the proposed approach is able to produce effective solutions for the problem instances.

The industrial contribution of the manuscript lies in its potential to optimize and enhance various aspects of home health care operations, leading to improved efficiency, cost savings, quality of care, improved the level of service, and resource utilization within the industry.

Moreover, the research originality hinges on introducing innovative concepts, methodologies, and problem-solving approaches that have the potential to significantly advance the field of home health care scheduling and routing. In addition, our manuscript focuses on a novel and previously unexplored aspect as a unique approach takes into consideration travel distances, stochastic demand, and other critical factors that have not been extensively studied in the existing literature.

An interesting direction for future research is related to the suggestion of specialized solution techniques to solve larger and highly constrained problem instances commonly existing in real-world scenarios. In addition, effective demand prediction is crucial to efficiently deal with stochastic scenarios.

On the other hand, the extension of the problem by the introduction of practical constraints can be an interesting direction for further work. For instance, flexible time windows of customers, patient preferences, and lunch-break time of nurses may present practical factors that can be integrated into the model. Furthermore, dynamic programming is also a promising extension in future work to avoid revising the overall scheduling in a stochastic environment such as the cancellation of a request visit or the provision of a new risky event.

Regarding the limitations of the study, we recognize the significance of solving realistic data instances as a crucial perspective that bridges the gap between theoretical advancements and practical applicability. The second limitation emphasizes the need for a more extensive comparative study. We understand that proving the efficiency and applicability of our proposed approach within a broader context is essential. To achieve this, a comprehensive comparative study with other relevant methods will be pivotal in validating the robustness and effectiveness of our approach.

Table 12

The results of the Wilcoxon-tests for the solutions achieved by the stochastic, deterministic, and robust models when solving R1 benchmark.

Indicators	V0		V1		V2		V3	
Models	D	R	D	R	D	R	D	R
+ Ranks	12	12	12	12	12	10	12	8
– Ranks	0	0	0	0	0	2	0	4
Test statistic	2.148	2.148	2.148	2.148	2.148	1.783	2.148	1.536
p-Value	0.001	0.001	0.001	0.001	0.001	0.005	0.001	0.037
Decision	Reject H0	Reject H0	Reject H0	Reject H0	Reject H0	Reject H0	Reject H0	Reject H0

Table 13

The results of the Wilcoxon-tests for the solutions achieved by the stochastic, deterministic, and robust models when solving C1 benchmark.

Indicators	V0		V1		V2		V3	
Models	D	R	D	R	D	R	D	R
+ Ranks	9	9	9	9	9	9	9	5
– Ranks	0	0	0	0	0	0	0	4
Test statistic	1.611	1.611	1.611	1.611	1.611	1.611	1.611	1.3251
p-Value	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.048
Decision	Reject H0	Reject H0	Reject H0	Reject H0	Reject H0	Reject H0	Reject H0	Reject H0

Table 14

The results of the Wilcoxon-tests for the solutions achieved by the stochastic, deterministic, and robust models when solving RC1 benchmark.

Indicators	V0		V1		V2		V3	
Models	D	R	D	R	D	R	D	R
+ Ranks	8	8	8	8	8	8	8	6
– Ranks	0	0	0	0	0	0	0	2
Test statistic	1.432	1.432	1.432	1.432	1.432	1.432	1.432	1.243
p-Value	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.0039
Decision	Reject H0	Reject H0	Reject H0	Reject H0	Reject H0	Reject H0	Reject H0	Reject H0

Table 15

The results of the Wilcoxon-tests for the solutions achieved by the stochastic, deterministic, and robust models when solving R2 benchmark.

Indicators	V0		V1		V2		V3	
Models	D	R	D	R	D	R	D	R
+ Ranks	11	11	11	10	11	1	11	10
– Ranks	0	0	0	1	0	3	0	1
Test statistic	1.969	1.969	1.969	1.969	1.969	1.969	1.969	1.969
p-Value	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.67
Decision	Reject H0	Reject H0	Reject H0	Reject H0	Reject H0	Retain H0	Reject H0	Retain H0

Table 16

The results of the Wilcoxon-tests for the solutions achieved by the stochastic, deterministic, and robust models when solving C2 benchmark.

Indicators	V0		V1		V2		V3	
Models	D	R	D	R	D	R	D	R
+ Ranks	8	8	0	0	0	0	0	0
– Ranks	0	0	8	0	0	0	0	2
Test statistic	1.432	1.432	1.432	–	–	–	–	–
p-Value	0.003	0.003	0.003	–	–	–	–	–
Decision	Reject H0	Reject H0	Reject H0	–	–	–	–	–

Table 17
The results of the Wilcoxon-tests for the solutions achieved by the stochastic, deterministic, and robust models when solving RC2 benchmark.

Indicators	V0		V1		V2		V3	
Models	D	R	D	R	D	R	D	R
+ Ranks	8	8	8	4	0	0	0	0
– Ranks	0	0	0	4	0	0	0	0
Test statistic	1.432	1.432	1.432	1.852	–	–	–	–
p-Value	0.003	0.003	0.003	0.07	–	–	–	–
Decision	Reject H0	Reject H0	Reject H0	Retain H0	–	–	–	–

CRediT authorship contribution statement

Marouene Chaieb: Conceptualization, Methodology, Software, Data curation, Formal analysis, Writing – original draft, Revise the manuscript. **Dhekra Ben Sassi:** Conceptualization, Software, Visualization, Revise the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

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